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CELL BIOLOGY

A hypoxia-responsive tRNA-derived small RNA confers renal protection through RNA autophagy

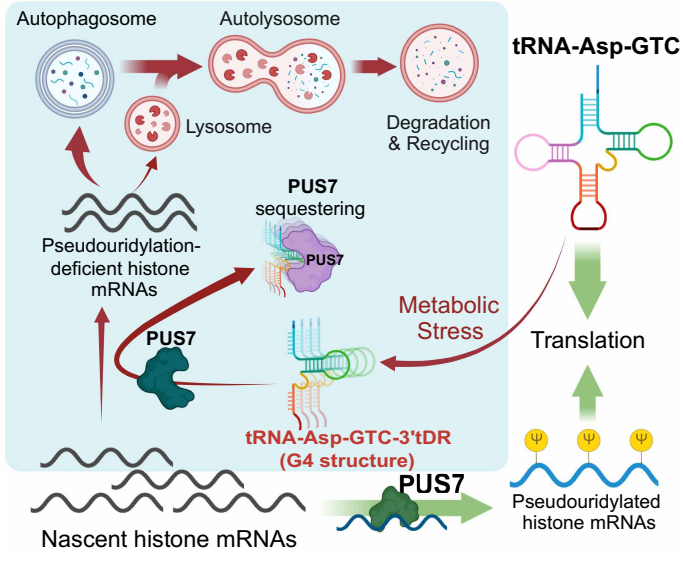
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Full article and list of author affiliations: https://doi.org/10.1126/science.adp5384

INTRODUCTION: Cleavage products of transfer RNAs (tRNAs) called tRNA-derived small RNAs (tsRNAs or tDRs), are an evolutionarily conserved class of noncoding small RNAs that play important roles in cellular stress responses. Derived from cleavage by endonucleases such as angiogenin at the anticodon loop, the 5′ halves of tRNA-Ala and tRNA-Cys have been implicated in the regulation of stress granules and global translation. Two tDRs derived from each end of tRNA-Asp-GTC, tRNA-Asp-GTC-3′tDR and tRNA-Asp-GTC-5′tDR, have been shown to be the most highly up-regulated hypoxia-responsive tDRs in multiple cell types. However, their functions in regulating the cellular stress response are not known.

RATIONALE: Although 5′tDRs have been extensively investigated, the regulation and function of 3′tDRs remain poorly characterized. We observed significant levels of tRNA-Asp-GTC-3′tDR in metabolically active tissues, notably the kidney. Validated murine models of kidney diseases and human tissue samples allowed us to investigate the regulation of this tDR in disease, as well as its clinical relevance. Reagents that we developed to modulate the levels of tRNA-Asp-GTC-3′tDR in cells and in vivo allowed us to determine the function of this tDR in kidney disease pathogenesis at a detailed molecular level.

RESULTS: We validated the biogenesis of hypoxia-induced tRNA-Asp-GTC-3′tDR in human embryonic kidney (HEK) cells using ultrasensitive Northern blotting and an engineered fluorescent reporter for tRNA-Asp-GTC-3′tDR, and further confirmed a critical role for the endonuclease angiogenin in its biogenesis. Using overexpression (with synthetic mimics) or specific silencing (using locked nucleic acid–modified antisense oligonucleotides, ASOs), we demonstrated that tRNA-Asp-GTC-3′tDR (but not the 5′tDR) was necessary and sufficient for driving autophagic flux in HEK cells. We found high basal levels of this tDR in most primary kidney cells, correlating with high basal autophagy in these cells. Silencing this tDR in primary kidney cells led to an inhibition of autophagy and cell death, suggesting a homeostatic role for tRNA-Asp-GTC-3′tDR. In several murine kidney disease models, as well as in human tissue samples, tRNA-Asp-GTC-3′tDR was rapidly up-regulated early after kidney injury. Using in vivo delivery of our ASOs, we demonstrated that silencing of tRNA-Asp-GTC-3′tDR led to inhibited autophagy, exacerbated kidney cell injury, and increased inflammation and fibrosis in two different kidney disease models, supporting a compensatory role for this tDR. Conversely, increasing the levels of this tDR using polymer nanoparticle–based delivery of the synthetic mimic was renoprotective, maintaining autophagy and markedly decreasing markers of kidney injury, inflammation, and fibrosis. Mechanistically, the regulation of autophagic flux by tRNA-Asp-GTC-3′tDR was critically dependent on binding to the RNA-modifying enzyme pseudouridine synthase 7 (PUS7). This interaction between tRNA-Asp-GTC-3′tDR and PUS7 relied on both structural oligo-guanine motifs, which allowed for stable intermolecular G-quadruplex (G4) structure formation, and a binding motif for PUS7. Binding and sequestration of PUS7 by tRNA-Asp-GTC-3′tDR prevented the pseudouridylation of target mRNAs, notably histone mRNAs. The pseudouridine-deficient histone mRNAs were targeted to the autophagosome-lysosome pathway for degradation, triggering RNA autophagy as a likely stress-adaptive response.



tRNA-Asp-GTC-3′tDR induces RNA autophagy in kidney cells. Under metabolic stress, kidney cells activate endonucleases such as angiogenin to cleave tRNA-Asp-GTC and generate tRNA-Asp-GTC-3′tDR. tRNA-Asp-GTC-3′tDRs form intermolecular G4 structures and sequester PUS7, thus preventing histone mRNA pseudouridylation. The resulting pseudouridine-deficient histone mRNAs entered the autophagosome-lysosome pathway, triggering RNA autophagy. This process serves as a protective stress response in kidney cells. [Figure created with Biorender.com]

CONCLUSION: This work adds to emerging studies on the role of tDRs in cellular homeostasis and the stress response. We found that the hypoxia-responsive tRNA-Asp-GTC-3′tDR maintains cellular homeostasis in kidney cells by regulating autophagic flux and plays a key role in the stress response. The levels of tRNA-Asp-GTC-3′tDR increased acutely in murine and human kidney diseases to enhance autophagic flux and protect against cellular injury, inflammation, and fibrosis. Mechanistically, key structural motifs in tRNA-Asp-GTC-3′tDR, including the oligo-guanine motif and T-arm region, are essential for both tDR stability and sequestration of the RNA-modifying enzyme PUS7. The sequestration subsequently drives RNA autophagy by preventing the pseudouridylation and stabilization of PUS7-targeted histone mRNAs. Targeting tRNA-Asp-GTC-3′tDR to maintain autophagic flux in the setting of kidney diseases may thus represent a promising therapeutic strategy. □

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CELL BIOLOGY

A hypoxia-responsive tRNA-derived small RNA confers renal protection through RNA autophagy

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Transfer RNA–derived small RNAs (tsRNAs or tDRs) perform a range of cellular functions. Here, we showed that tRNA-Asp-GTC-3′tDR, a hypoxia-induced tDR derived from the 3′ end of tRNA-Asp-GTC, activated autophagic flux in kidney cells and its silencing blocked autophagic flux. Functional gain-/loss-of-function studies in murine kidney disease models demonstrated a substantial renoprotective function of tRNA-Asp-GTC-3′tDR. Mechanistically, tRNA-Asp-GTC-3′tDR assembled stable G-quadruplex structures and sequestered pseudouridine synthase 7 (PUS7), preventing catalytic pseudouridylation of histone mRNAs. The resulting pseudouridylation deficiency directed histone mRNAs to the autophagosome-lysosome pathway, triggering RNA autophagy. This tDR-induced RNA autophagy pathway was activated during murine and human kidney diseases, suggesting clinical relevance. Thus, tRNA-Asp-GTC-3′tDR plays a role in regulating RNA autophagy, which helps to maintain homeostasis in kidney cells and protects against kidney injury.

Transfer RNAs (tRNAs) act as amino acid carriers and mRNA decoders during protein synthesis (*1*). The human genome contains >500 tRNA genes, with ~50% actively transcribing tRNAs, making them the most abundant RNA species within cells (*2*, *3*). Certain tRNAs can be cleaved into fragments of various lengths called tRNA-derived small RNAs (tsRNAs or tDRs), particularly as a response to stress conditions (*4*). The development of specialized RNA-sequencing methodologies has accelerated the recognition of a plethora of tDRs (*5*–*8*). We now understand that tRNA cleavage represents an evolutionarily conserved process that produces new RNA species in all domains of life (*9*). The biogenesis of tDRs is orchestrated by various ribonucleases, such as DICER, RNaseL, and angiogenin (ANG), together with RNA-binding proteins (RBPs) and RNA modification enzymes, rendering it sensitive to fluctuating cellular dynamics (*10*).

The intricate regulation of tDR biogenesis, along with their unique linear sequences and complex three-dimensional structures, underscore the multifaceted roles of tDRs as an emerging class of regulatory small noncoding RNAs. tDR species have been implicated in a myriad of biological processes (*11*), including regulation of translation and transcription (*12*–*17*), RNA interference–like functions (*18*), and regulation of mRNA stability through competitive binding of RBPs (*19*).

From a comprehensive tDR atlas generated using AlkB-facilitated RNA methylation sequencing (ARM-seq) to characterize stress-regulated tDRs across multiple cell types, we identified >3000 significantly regulated hypoxia-responsive tDRs (*20*). In this work, we focused on tRNA-Asp-GTC-3′tDR, derived from the 3′ end of tRNA-Asp-GTC, as the most up-regulated tDR in response to hypoxia and studied its biogenesis and regulation. Using silencing and overexpression tools, we explored the functional role of tRNA-Asp-GTC-3′tDR in regulating cellular stress responses in kidney cells. We next investigated its role in the pathogenesis of kidney injury in two different kidney disease models using complementary gain-of-function (GOF) and loss-of-function (LOF) approaches. Finally, we characterized the key structural motifs and binding partners of tRNA-Asp-GTC-3′tDR that were the critical determinants of its function in kidney cells.

Hypoxia-induced tRNA-Asp-GTC-3′tDR enhances autophagic flux

In a comprehensive stress-specific cellular and extracellular tDR atlas (*20*), analysis of multiple cell types identified numerous tDRs highly regulated by different stressors. To characterize their functions, we focused on tDRs demonstrating the most significant stress-induced changes in human embryonic kidney (HEK) cells. ARM-seq data showed that hypoxia induced the most substantial changes in tDR levels in HEK cells (fig. S1A). The top two tDRs up-regulated by hypoxia originated from the tRNA-Asp-GTC-2 gene, with one from the 5′ end [named by tDRnamer (*21*) as tDR-1:32-Asp-GTC-2; hereafter tRNA-Asp-GTC-5′tDR] and another one from the 3′ end (named by tDRnamer as tDR-39:72-Asp-GTC-2-M2; hereafter tRNA-Asp-GTC-3′tDR) (Fig. 1A and table S1). Northern blotting validated their up-regulation, demonstrating additionally that tDR isoforms comprise <3% of the full-length tRNA-Asp-GTC (Fig. 1B and fig. S1, B and C). To confirm that these tDRs were generated by tRNA cleavage, we designed a self-quenched tDR biogenesis reporter. We conjugated a fluorescent group (fluorescein isothiocyanate, FITC) and a quencher group (black hole quencher 1, BHQ-1) to each end of full-length tRNA-Asp-GTC (Fig. 1C), which only releases fluorescent signals upon tRNA cleavage and separation of 5′tDR and 3′tDR (*22*). HEK cells transfected with this reporter exhibited significantly higher FITC signals under hypoxic conditions compared with normoxic conditions (Fig. 1, D and E), confirming hypoxia-induced tRNA-Asp-GTC fragmentation and tDR biogenesis. Previous studies identified ANG as a stress-activated ribonuclease that generates tDRs by cleaving tRNAs at the anticodon loop (*23*, *24*). Recent work identified ANG as one of the most up-regulated ribonucleases under hypoxia (*20*). To determine the role of ANG in tRNA-Asp-GTC-3′tDR biogenesis, we subjected an ANG-deleted cell line (*20*) to hypoxia treatment and evaluated the levels of tRNA-Asp-GTC-3′tDR.

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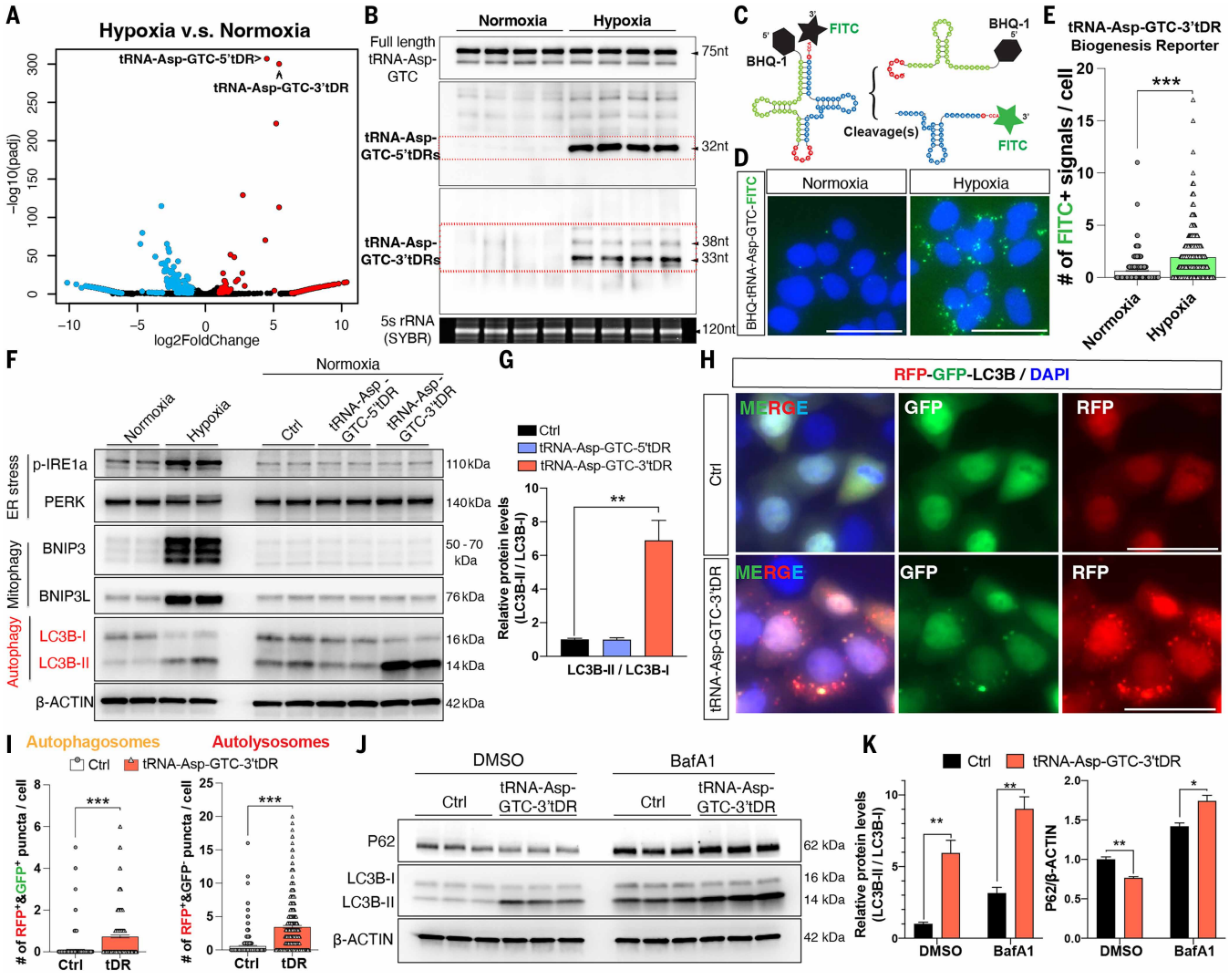


Fig. 1. Hypoxia-induced tRNA-Asp-GTC-3′tDR enhances autophagic flux. (**A**) Volcano plot showing significantly regulated tDRs in HEK cells upon hypoxia treatment, with the top two significantly up-regulated tDRs labeled. (**B**) Northern blotting validating the up-regulation of 5′tDR and 3′tDR of tRNA-Asp-GTC in HEK cells upon hypoxia. (**C**) Schematic illustration of tDR biogenesis reporter design. (**D**) Live imaging of fluorescent signals from the tRNA-Asp-GTC-3′tDR biogenesis reporter in HEK cells with or without hypoxia treatment. (**E**) Quantification of FITC-positive signals per cell from (D). (**F**) Immunoblot analysis of different protein markers involved in ER stress, mitophagy, and autophagy pathways in HEK cells transfected with control (Ctrl) and tRNA-Asp-GTC-5′tDR and tRNA-Asp-GTC-3′tDR mimics. (**G**) Quantification of protein levels in (F). (**H**) Live imaging of fluorescent signals in RFP-GFP-LC3B autophagy reporter HEK cells transfected with Ctrl or tRNA-Asp-GTC-3′tDR mimics. (**I**) Quantification of the number of autophagosomes (GFP^+ and RFP^+) and autolysosomes (GFP^- and RFP^+) per cell in (H). (**J**) Immunoblot analysis of autophagic protein levels in HEK cells transfected with Ctrl or tRNA-Asp-GTC-3′tDR mimics and treated with dimethyl sulfoxide (DMSO) or BafA1 for 6 hours. (**K**) Quantification of protein levels in (J). Data are shown as means ± SEM of at least three independent experiments. Mann-Whitney test was used in (E) and (I). Unpaired two-tailed Student's *t* test was used in (G) and (K). $*P < 0.05$, $**P < 0.01$, $***P < 0.001$. Scale bar, 50 μm.

ANG deletion substantially impaired hypoxia-induced generation of tRNA-Asp-GTC-3′tDR (fig. S1, D to F), establishing ANG as a critical ribonuclease for tRNA-Asp-GTC-3′tDR biogenesis under hypoxic conditions, as expected (*25*). Although ANG cleavage appears to be the dominant mechanism for tRNA-Asp-GTC-3′tDR generation, the residual levels of this tDR in ANG-deleted cells suggest that other effectors may also be involved in its biogenesis.

Hypoxia triggers multiple cellular stress responses, including endoplasmic reticulum (ER) stress, mitophagy, and autophagy, and tDRs have previously been shown to induce translational repression and stress granule formation during cellular stress (*12*–*15*, *26*–*28*). To investigate the role of hypoxia-induced tDRs in the hypoxic stress response, we transfected synthetic RNA oligos of tRNA-Asp-GTC-5′tDR or tRNA-Asp-GTC-3′tDR into HEK cells and analyzed the resulting protein lysates. Immunoblotting demonstrated that hypoxia activated ER stress (phosphorylation of IRE1a and down-regulation of PERK), mitophagy (induction of BNIP3 and BNIP3L), and autophagy (conversion of LC3B-I to lipidated LC3B-II) (Fig. 1F). Neither tDR affected ER stress or mitophagy pathways. However, introduction of tRNA-Asp-GTC-3′tDR, but not tRNA-Asp-GTC-5′tDR, led to the activation of autophagy pathways under normoxic conditions (Fig. 1, F and G). Using a GFP-LC3B autophagy reporter cell line, we confirmed that tRNA-Asp-GTC-3′tDR significantly promoted the formation of green fluorescent protein (GFP)–positive punctate-like structures within hours of transfection (fig. S1, G to J).

Autophagic flux encompasses autophagosome formation, autophagosome-lysosome fusion, and substrate degradation in autolysosomes (*29*). To determine how tRNA-Asp-GTC-3′tDR regulates autophagic flux, we built an autophagic flux reporter cell line expressing red fluorescent protein (RFP)–GFP tandem fluorescent-tagged LC3B.

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RFP-GFP-LC3B–labeled autophagosomes show both GFP and RFP signals before fusion with lysosomes and only RFP signals after fusion (*29*). tRNA-Asp-GTC-3′tDR promoted the formation of both autophagosomes (GFP^+/RFP^+) and autolysosomes (GFP^-/RFP^+) (Fig. 1, H and I), indicating enhanced autophagic flux. Autophagosome-lysosome fusion is usually associated with activated lysosomal activity (*30*), and indeed we observed increased lysosomal activity in tRNA-Asp-GTC-3′tDR–transfected cells (fig. S1, K and L). We used bafilomycin A1 (BafA1), which inhibits autophagosome maturation, to measure autophagic flux (*29*, *31*). After BafA1 treatment, the accumulation of lipidated LC3B and P62 protein levels was significantly higher in tRNA-Asp-GTC-3′tDR–transfected cells compared with controls (Fig. 1, J and K). Finally, additionally boosting tRNA-Asp-GTC-3′tDR levels in hypoxic HEK cells further enhanced autophagic flux (fig. S1, M and N). Thus, tRNA-Asp-GTC-3′tDR enhances autophagic flux.

Inhibition of tRNA-Asp-GTC-3′tDR blocks autophagic flux in different kidney cell types

Synthetic locked, nucleic acid–modified antisense oligonucleotides (ASOs) have been used to specifically block tDR function (*19*, *32*). We thus adopted the ASO inhibition approach to further elucidate the function of endogenous tRNA-Asp-GTC-3′tDR. After screening several designed ASOs, we identified one, Asp-GTC-3′tDR-ASO (fig. S2A), that could efficiently knock down tRNA-Asp-GTC-3′tDR levels in hypoxic HEK cells with no significant change in the parent tRNA-Asp-GTC expression compared with negative control ASO (NC-ASO) (Fig. 2, A and B). The targeting specificity was further confirmed by a computational tube reaction model, which incorporates four components at equal concentrations (1 μM), Asp-GTC-3′tDR-ASO, tRNA-Asp-GTC-3′tDR, tRNA-Asp-GTC-5′tDR, and full-length tRNA-Asp-GTC, and accounts for their respective secondary structures. On the basis of the calculated equilibrium complex concentrations, the 3′tDR+ASO hybrid emerged as the most dominant species, reaching a concentration of 0.91 μM, whereas the ASO+full-length tRNA hybrid achieved only minimal formation, with a concentration of 0.086 μM (fig. S2, A and B). Thus, >90% of the Asp-GTC-3′tDR-ASO preferentially binds to the 3′tDR, whereas <1% interacts with full-length tRNAs.

We next assayed the effects of tRNA-Asp-GTC-3′tDR knockdown on autophagy pathways. The inhibition of tRNA-Asp-GTC-3′tDR using Asp-GTC-3′tDR-ASO significantly increased LC3B-II protein levels in HEK cells under both normoxic and hypoxic conditions (Fig. 2, C and D), suggesting either increased autophagy initiation or a block in

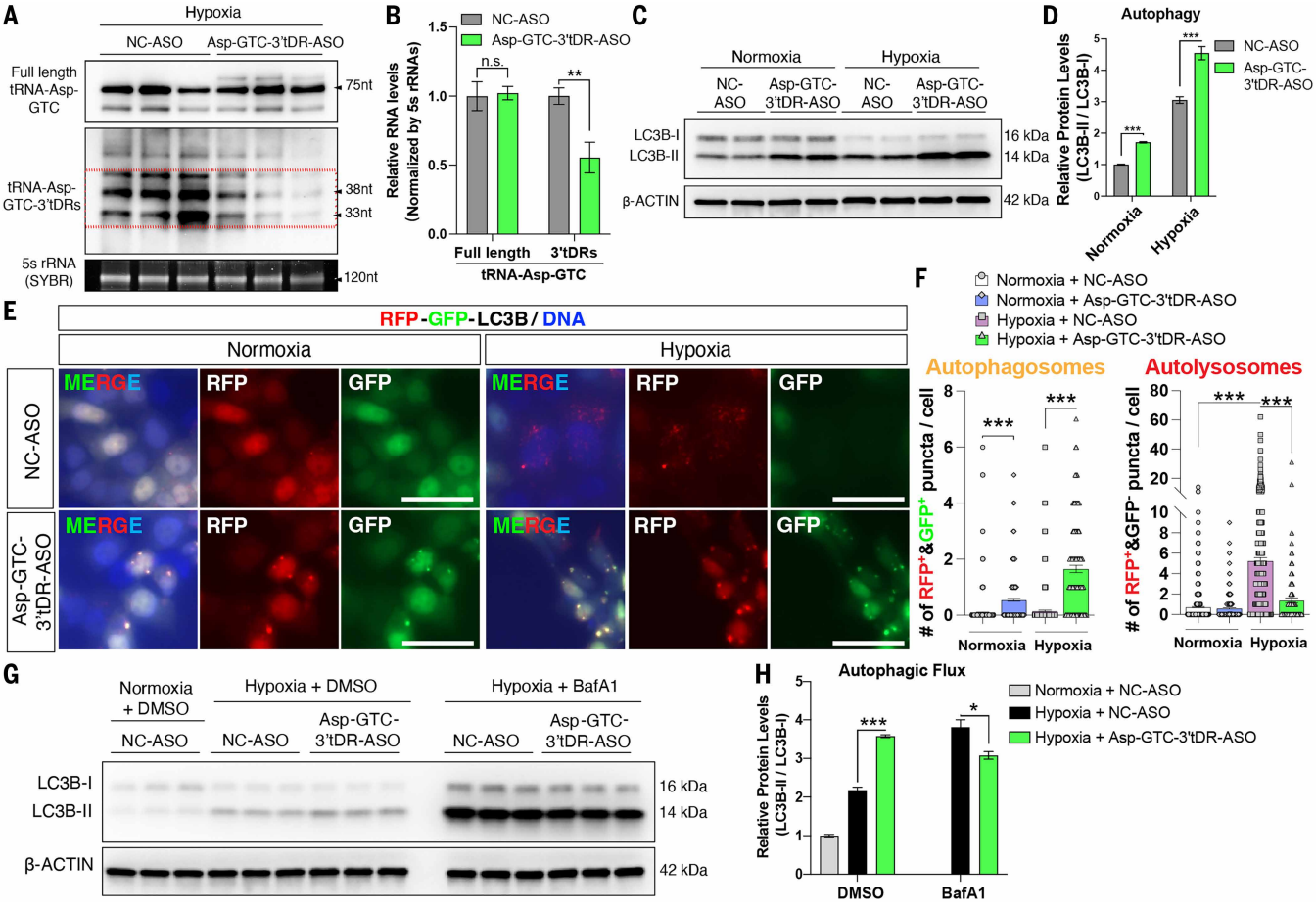


Fig. 2. Inhibition of tRNA-Asp-GTC-3′tDR blocks autophagic flux. (**A**) Northern blotting showing the levels of full-length and 3′tDR of tRNA-Asp-GTC in hypoxic HEK cells transfected with negative Ctrl ASO (NC-ASO) or tRNA-Asp-GTC-3′tDR ASO (Asp-GTC-3′tDR-ASO). (**B**) Quantification of levels of full-length and 3′tDR of tRNA-Asp-GTC in (A). (**C**) Immunoblot analysis of protein levels of LC3B and β-actin in HEK cells transfected with Ctrl or tRNA-Asp-GTC-3′tDR ASOs under normoxic or hypoxic conditions. (**D**) Quantification of protein levels in (C). (**E**) Live imaging of fluorescent signals in RFP-GFP-LC3B autophagy reporter HEK cells transfected with NC-ASO or Asp-GTC-3′tDR-ASO under normoxic or hypoxic conditions. (**F**) Quantification of the number of autophagosomes (GFP^+ and RFP^+) and autolysosomes (GFP^- and RFP^+) per cell in (E). (**G**) Immunoblot analysis of protein levels of LC3B and β-actin in hypoxic HEK cells transfected with NC-ASO or Asp-GTC-3′tDR-ASO and treated with DMSO or BafA1 for 6 hours. (**H**) Quantification of protein levels in (G). Data are shown as means ± SEM of at least three independent experiments. Unpaired two-tailed Student's *t* test was used in (B), (D), and (H). Mann-Whitney test was used in (F). n.s., nonsignificant, $*P < 0.05$, $**P < 0.01$, $***P < 0.001$. Scale bar, 50 μm.

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autophagic flux. Using the RFP-GFP-LC3B reporter cell line, we found that Asp-GTC-3′tDR-ASO treatment resulted in autophagosome accumulation without maturation into autolysosomes under both normoxic and hypoxic conditions (Fig. 2, E and F), aligning with increased LC3B-II protein levels. Treatment with BafA1 caused further LC3B-II accumulation in both groups but with significantly lower levels in tRNA-Asp-GTC-3′tDR–inhibited cells under hypoxia compared with controls (Fig. 2, G and H). Thus, the increased LC3B-II levels noted above resulted from a block in autophagic flux caused by tRNA-Asp-GTC-3′tDR inhibition.

To characterize the physiological roles of tRNA-Asp-GTC-3′tDR, we analyzed its tissue abundance through ARM-seq of different mouse tissues and observed the highest abundance in kidney, heart, and brain tissues, particularly of a 38-nt isoform, tDR-38:76-Asp-GTC-2-M2 (Fig. 3A). We tested the functionality of this mouse 38-nt isoform and another abundant 34-nt isoform, tDR-38:72-Asp-GTC-2-M2, which was detected by ARM-seq and Northern blotting (fig. S2C). As expected, both isoforms activated autophagy flux like the 33-nt isoform (fig. S2, D and E). We also detected abundant levels of tRNA-Asp-GTC-3′tDRs, especially the 38-nt isoform, in a variety of human kidney cells, including proximal tubular epithelial cells (PTEpCs), kidney fibroblasts, glomerular microvascular endothelial cells (GMECs), glomerular epithelial cells (GEpCs), podocytes, and glomerular mesangial cells (Fig. 3B and fig. S3A). Healthy kidney cells, especially PTEpCs and podocytes, maintain high basal autophagy levels, which are essential for maintaining kidney homeostasis (*33*, *34*). Our analysis of autophagy activity and tRNA-Asp-GTC-3′tDR abundance suggested that high tRNA-Asp-GTC-3′tDR levels may also sustain high basal autophagic flux in these

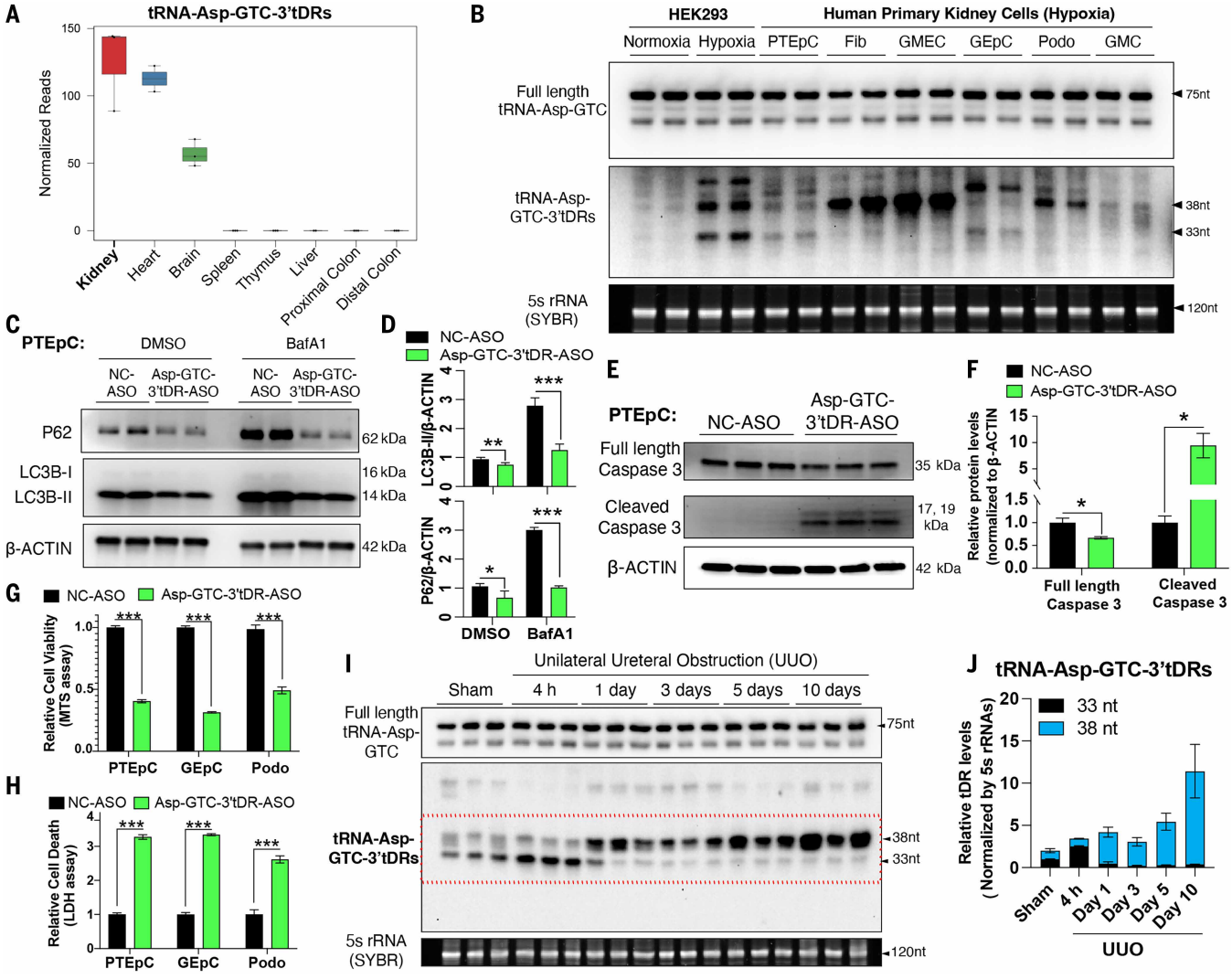


Fig. 3. tRNA-Asp-GTC-3′tDR sustains autophagy flux in different kidney cell types. (**A**) ARM-seq analysis showing the abundance of tRNA-Asp-GTC-3′tDR across mouse organs. (**B**) Northern blotting showing the levels of full-length and 3′tDR of tRNA-Asp-GTC across different human kidney cells under hypoxia conditions. Fib, primary kidney fibroblasts; Podo, immortalized podocytes; GMC, primary glomerular mesangial cells. (**C**) Immunoblot analysis of autophagic protein levels in PTEpCs transfected with NC-ASO or Asp-GTC-3′tDR-ASO and treated with DMSO or BafA1 for 6 hours. (**D**) Quantification of protein levels in (C). (**E**) Immunoblot analysis of the protein levels of full-length or cleaved caspase 3 in human primary PTEpCs transfected with NC-ASO or Asp-GTC-3′tDR-ASO. (**F**) Quantification of full-length or cleaved caspase 3 protein levels in (E). (**G** and **H**) Cell viability assessment by MTS assay (G) and cell death assessment by secreted lactate dehydrogenase assay (H) in different kidney cells after transfection with NC-ASO or Asp-GTC-3′tDR-ASO. (**I**) Northern blotting showing the levels of full-length and 3′tDRs of tRNA-Asp-GTC in the obstructed kidney tissues at different time points after UUO surgery. (**J**) Quantification of the levels of tRNA-Asp-GTC-3′tDR isoforms (33 and 38 nt) in (I). Data are shown as means ± SEM of at least three independent experiments. Unpaired two-tailed Student's *t* test was used in (D), (F), (G), and (H). $*P < 0.05$, $**P < 0.01$, $***P < 0.001$.

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kidney cells (fig. S3, A to C). As expected, tRNA-Asp-GTC-3′tDR inhibition by Asp-GTC-3′tDR-ASO impeded overall autophagic flux in various human kidney cells, including PTEpCs, GEpCs, and GMECs (Fig. 3, C and D, and fig. S3, D to G). In contrast to HEK cells, in these kidney cells, inhibition of basal tRNA-Asp-GTC-3′tDR affected generalized autophagy flux, not just autophagosome maturation. Autophagy is essential for the function and survival of renal cells (*33*). Failure to sustain desired the autophagy flux in these kidney cells resulting from tRNA-Asp-GTC-3′tDR inhibition triggered apoptosis and cell death (Fig. 3, E to H, and fig. S3, H and I). Thus, tRNA-Asp-GTC-3′tDR is necessary for maintaining autophagic flux in kidney cells, which is indispensable for their health.

Inhibition of tRNA-Asp-GTC-3′tDR exacerbates kidney injuries in mouse models

To investigate the pathophysiological role of tRNA-Asp-GTC-3′tDR in vivo, we validated its basal abundance in kidneys and found marked increases after various kidney injuries, including uninephrectomy-induced volume and hemodynamic overload and albumin injection–induced protein overload (fig. S4, A to D). In the mouse unilateral ureteral obstruction (UUO) model, which resembles human chronic obstructive nephropathy (*35*), tRNA-Asp-GTC-3′tDR levels were promptly up-regulated in the obstructed kidney within 4 hours after surgery and remained elevated for 10 days (Fig. 3, I and J), with a switch from the 33- to the 38-nt isoform on day 1 after surgery. We hypothesized that this augmented biogenesis of tRNA-Asp-GTC-3′tDRs enhances kidney autophagic flux as an adaptive response to ameliorate the progression of kidney injury.

To investigate the role of tRNA-Asp-GTC-3′tDR in the UUO-induced kidney injury model, we used Asp-GTC-3′tDR-ASO with phosphorothioate backbones for enhanced stability and self-delivery (*36*). ASO was administered intravenously into mice every other day from the day before UUO surgery, with tissue pathology analysis on day 6 (Fig. 4A). The knockdown of tRNA-Asp-GTC-3′tDRs in the obstructed kidney was confirmed using Northern blotting (Fig. 4, B and C, and fig. S5A). Ureteral obstruction usually results in marked tubular injury and cell death, with interstitial immune cell infiltration and fibroblast activation (*35*). Consistent with this, UUO surgery led to acute kidney injury, immune cell infiltration, and fibrosis in the obstructed kidney from the NC-ASO–treated group (Fig. 4D). Asp-GTC-3′tDR-ASO administration further increased markers for kidney injury, immune cell infiltration, and fibrosis in the obstructed kidney compared with the NC-ASO group (Fig. 4D). This was corroborated by immunohistology, which revealed substantial increases in apoptosis, neutrophil and macrophage infiltration, and collagen deposition (Fig. 4, E to G) in the obstructed kidney after tRNA-Asp-GTC-3′tDR inhibition. Immunoblotting further validated a consistent phenotype of exacerbated kidney injury after tRNA-Asp-GTC-3′tDR knockdown (Fig. 4, H and I). tRNA-Asp-GTC-3′tDR knockdown inhibited autophagic flux in the obstructed kidney (Fig. 4, H and I). We also observed mild kidney injury after tRNA-Asp-GTC-3′tDR inhibition in the sham surgery group (Fig. 4D), which may have been due to disordered autophagy at baseline. To validate this further, we used a 4-week chronic knockdown approach by subcutaneously injecting ASOs weekly into healthy mice. Upon confirming the knockdown of tRNA-Asp-GTC-3′tDR (fig. S5, B and C), we examined the kidney tissues using transmission electron microscopy. Indeed, inhibition of tRNA-Asp-GTC-3′tDR impaired autophagosome maturation, as indicated by deficient autophagosome-lysosome fusion in kidney tissues (fig. S5D). Thus, tRNA-Asp-GTC-3′tDR abundance is increased across kidney injury models and may be part of a compensatory response to maintain autophagic flux, and its inhibition consequently exacerbates kidney injury.

Delivery of tRNA-Asp-GTC-3′tDR alleviates renal ischemia/reperfusion–induced kidney injury in mice

We further conducted GOF/LOF studies in a renal ischemia/reperfusion injury (IRI) model. tRNA-Asp-GTC-3′tDR levels initially increased 1 day after IRI but returned to baseline by day 7 (Fig. 5, A and B). Similar to the UUO model, inhibition of tRNA-Asp-GTC-3′tDR using Asp-GTC-3′tDR-ASO worsened kidney injury, immune cell infiltration, and fibrosis in the kidney after IRI (fig. S5E). We next investigated whether further elevating tRNA-Asp-GTC-3′tDR levels could boost autophagy and attenuate kidney injury in this IRI model. We evaluated two nonviral vehicles for tDR delivery and identified that polymer-based polyethylenimine (PEI) efficiently delivered synthetic tDR mimics to kidneys (fig. S6, A and B), which is consistent with previous studies (*37*, *38*). We administered tDR mimics using PEI in this IRI model (Fig. 5C). Northern blotting confirmed the successful kidney delivery of tRNA-Asp-GTC-3′tDR mimics (Fig. 5, D and E). During the typical course of renal IRI, acute tubular injury, cell death, and immune cell infiltration occur within the first 2 days, followed by gradual repair after day 7, whereas the resulting fibrosis shows only partial resolution (*39*). Delivery of tRNA-Asp-GTC-3′tDR mimics significantly alleviated IRI-induced kidney injury compared with the control mimic-treated group, as demonstrated by decreased expression of makers for tubular injury, immune cell infiltration, and fibrosis, along with increased expression of Klotho (Fig. 5, F and G). Immunohistology analysis showed reduced apoptosis and attenuated fibroblast activation and collagen deposition (Fig. 5, H to J). tRNA-Asp-GTC-3′tDR administration enhanced the autophagy pathway in both sham and IRI kidneys compared with the control group (Fig. 5, K and L). Thus, the tRNA-Asp-GTC-3′tDR–autophagy axis serves a crucial protective function in kidneys.

tRNA-Asp-GTC-3′tDR assembles stable intermolecular G-quadruplex structures

We next explored the molecular mechanism underlying the regulation of autophagic flux by tRNA-Asp-GTC-3′tDR, which is a guanine-rich RNA (~50%) containing two 4-guanine motifs (fig. S2C). Previous studies have reported G-quadruplex (G4) structure formation by two other 5′tDRs containing terminal oligo-guanine motifs (*40*). To examine potential G4 formation by tRNA-Asp-GTC-3′tDR, we used complementary analytical approaches. We first assembled tRNA-Asp-GTC-3′tDR mimics under G4-permissive (K^+) or G4-nonpermissive (Li^+) conditions and analyzed assemblies using native gel electrophoresis. Under G4-permissive conditions, tRNA-Asp-GTC-3′tDR formed multiple intermolecular assemblies that were absent under G4-nonpermissive conditions (Fig. 6A). Circular dichroism (CD) spectroscopy revealed characteristic G4 structural signatures: a monovalent cation-responsive peak at ~263 nm (high intensity in a K^+ environment and low intensity in a Li^+ environment) and a trough at ~240 nm (Fig. 6B). Finally, we stained these tDR assemblies with *N*-methylmesoporphyrin (NMM), a G4-specific fluorescent probe (*41*), and found that tRNA-Asp-GTC-3′tDR formed various NMM-positive multimolecular G4 structures only under the G4-permissive conditions (Fig. 6C). We also confirmed the G4-forming ability of the 38-nt isoform through electrophoretic gel mobility, CD spectrum, and thermostability analyses (fig. S7, A to C). Thus, the different isoforms of tRNA-Asp-GTC-3′tDR assemble intermolecular G4 structures.

To identify the role of 4-guanine motifs in G4 formation, we generated variants with two 4-guanine motifs mutated individually (the 1GU and 2GU mutants) or simultaneously (the 12GU mutant) by replacing the second guanine with uridine in each motif. Comprehensive biochemical analyses demonstrated that the 1GU and 2GU mutants exhibited decreased capacities to form G4 structures, whereas the 12GU mutant completely abolished G4 structure assembly (Fig. 6, A to C, and fig. S7D). G4 structure assembly significantly enhanced its molecular stability, as evidenced by the detection of intact tRNA-Asp-GTC-3′tDR mimics after a 16-hour incubation with crude cell lysate (Fig. 6D). By contrast, 4-guanine motif mutants assembled under G4-permissive conditions, control mimics, and tDR mimics assembled under the G4-nonpermissive condition all failed to maintain their

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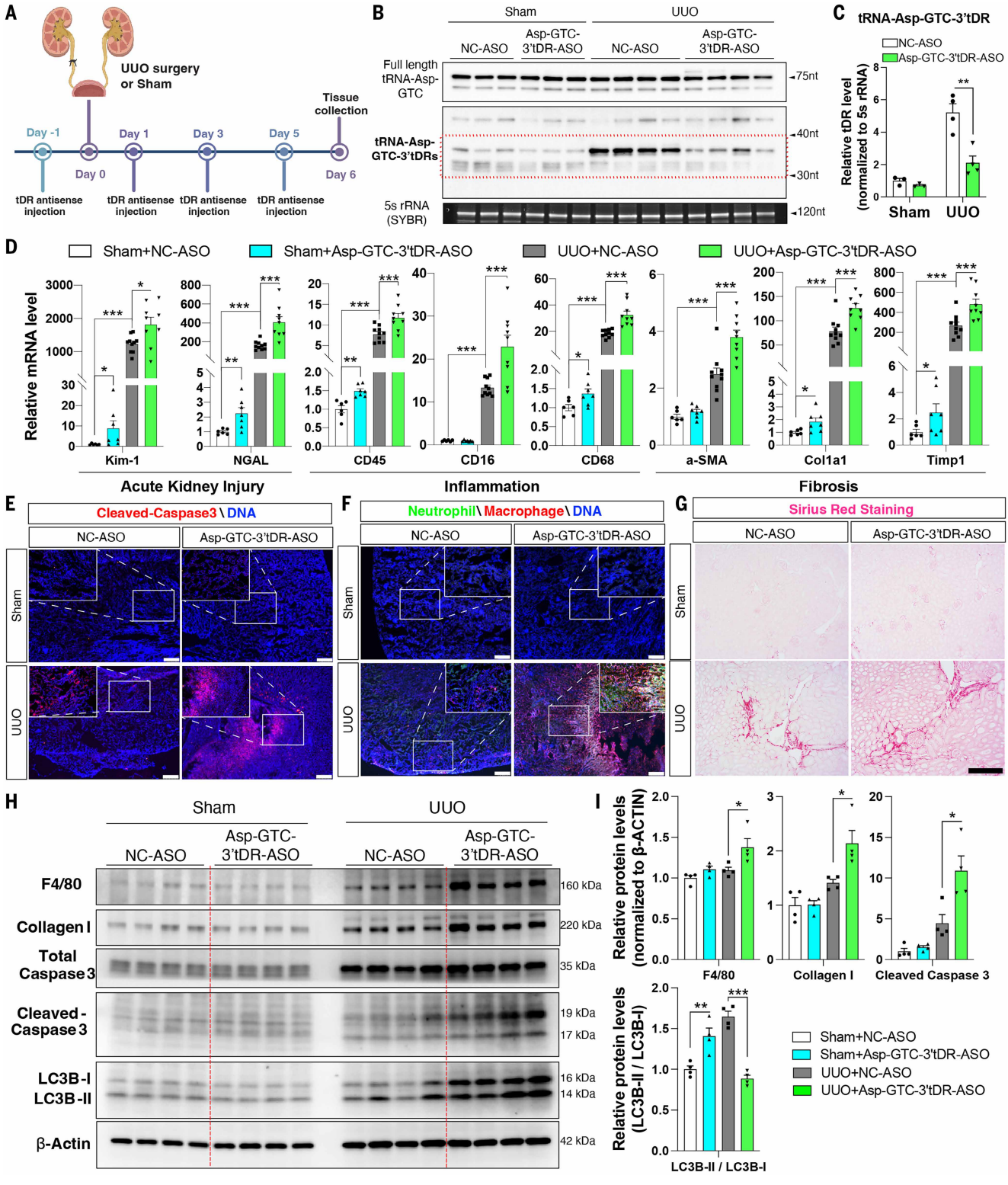


Fig. 4. Inhibition of tRNA-Asp-GTC-3′tDR exacerbates kidney injuries in mouse UUO models. (**A**) Schematic illustration of experimental design for tRNA-Asp-GTC-3′tDR inhibition in the mouse UUO model. (**B**) Northern blotting showing the levels of full-length and 3′tDR of tRNA-Asp-GTC in kidney tissues from sham or UUO (obstructed kidney) mice treated with NC-ASO or Asp-GTC-3′tDR-ASO. (**C**) Quantification of levels of tRNA-Asp-GTC-3′tDR in (B). (**D**) Analysis of gene expression related to acute kidney injury, immune cell infiltration, and fibrosis in kidney tissues from sham or UUO mice treated with NC-ASO or Asp-GTC-3′tDR-ASO ($n = 6$ to 10 per group). (**E** and **F**) Immunostaining analysis of cleaved-caspase 3 (E) and Ly6B.2 (neutrophil) and F4/80 (macrophage) (F) in kidney tissues from sham or UUO mice treated with NC-ASO or Asp-GTC-3′tDR-ASO. (**G**) Sirius red staining analysis of collagen levels in kidney tissues from sham or UUO mice treated with NC-ASO or Asp-GTC-3′tDR-ASO. (**H**) Immunoblot analysis of protein markers associated with immune cells, fibrosis, apoptosis, and autophagy in kidney tissues from sham or UUO mice treated with NC-ASO or Asp-GTC-3′tDR-ASO. (**I**) Quantification of protein levels in (H). Data are shown as means ± SEM of at least three independent experiments. Unpaired two-tailed Student's *t* test was used in (C) and (I). Mann-Whitney test was used in (D). $*P < 0.05$, $**P < 0.01$, $***P < 0.001$. Scale bar, 250 μm.

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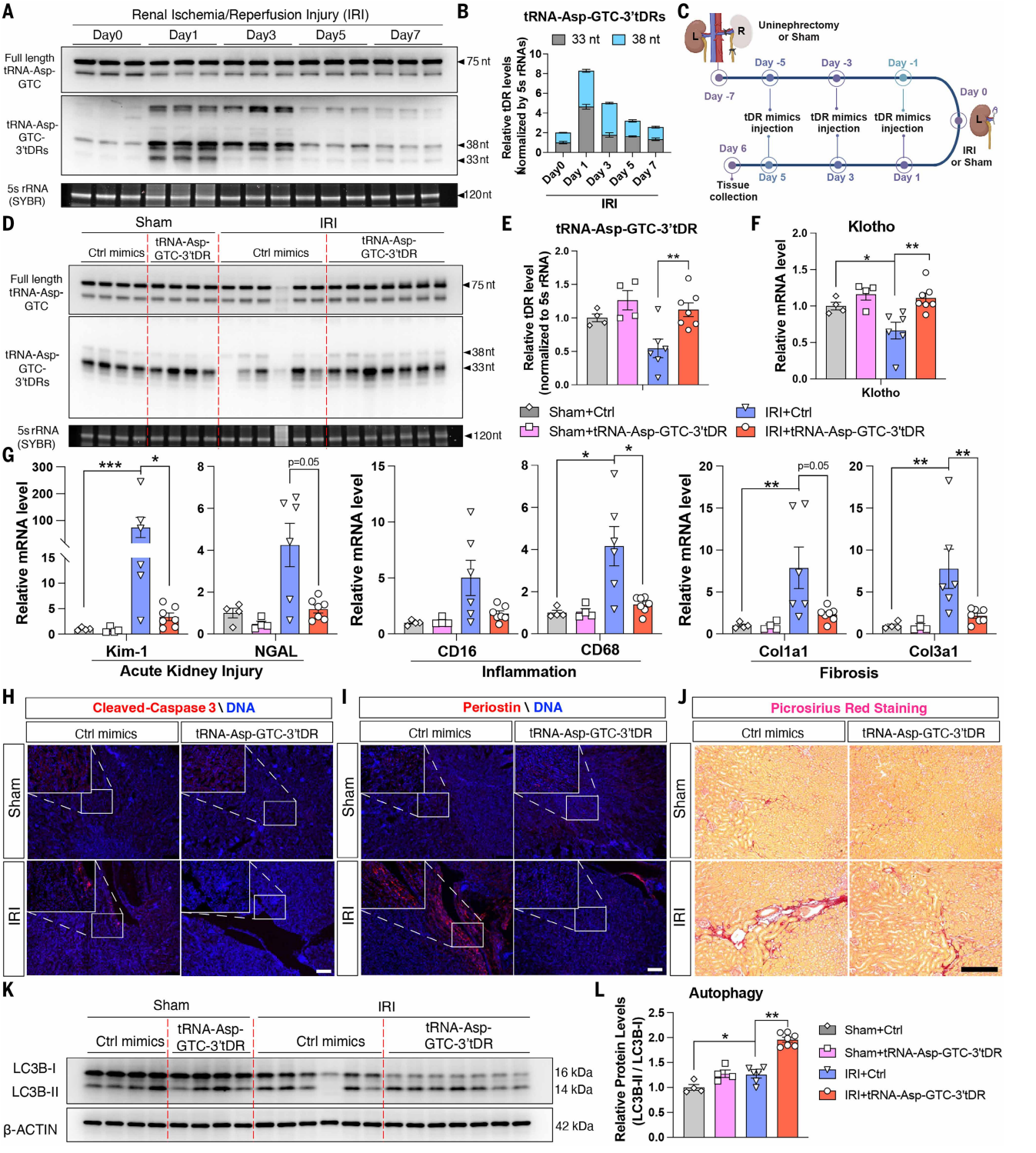


Fig. 5. Delivery of tRNA-Asp-GTC-3′tDR mimics protects against renal IRI in mice. (**A**) Northern blotting showing the levels of full-length and 3′tDR of tRNA-Asp-GTC in the kidney from the mouse renal IRI model at different time points. (**B**) Quantification of the levels of full-length and 3′tDR of tRNA-Asp-GTC in (A). (**C**) Schematic illustration of the experimental design for tRNA-Asp-GTC-3′tDR delivery in the mouse renal IRI model. (**D**) Northern blotting showing the levels of full-length and 3′tDR of tRNA-Asp-GTC in kidney tissues from sham or IRI mice treated with Ctrl or tRNA-Asp-GTC-3′tDR mimics. (**E**) Quantification of tRNA-Asp-GTC-3′tDR levels in (D). (**F**) Analysis of Klotho expression in kidney tissues from sham or IRI mice treated with Ctrl or tRNA-Asp-GTC-3′tDR mimics. (**G**) Analysis of gene expression related to acute kidney injury, immune cell infiltration, and fibrosis in kidney tissues from sham or IRI mice treated with Ctrl or tRNA-Asp-GTC-3′tDR mimics ($n = 4$ to 7 per group). (**H** and **I**) Immunostaining analysis of cleaved-caspase 3 (H) and periostin (I) in kidney tissues from sham or IRI mice treated with Ctrl or tRNA-Asp-GTC-3′tDR mimics. (**J**) Picrosirius red staining analysis of collagen levels in kidney tissues from sham or IRI mice treated with Ctrl or tRNA-Asp-GTC-3′tDR mimics. (**K**) Immunoblot analysis of the autophagy pathway in kidney tissues from sham or IRI mice treated with Ctrl or tRNA-Asp-GTC-3′tDR mimics. (**L**) Quantification of protein levels in (K). Data are shown as means ± SEM of at least three independent experiments. Mann-Whitney test was used in (E), (F), (G), and (L). $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$. Scale bar, 250 μm.

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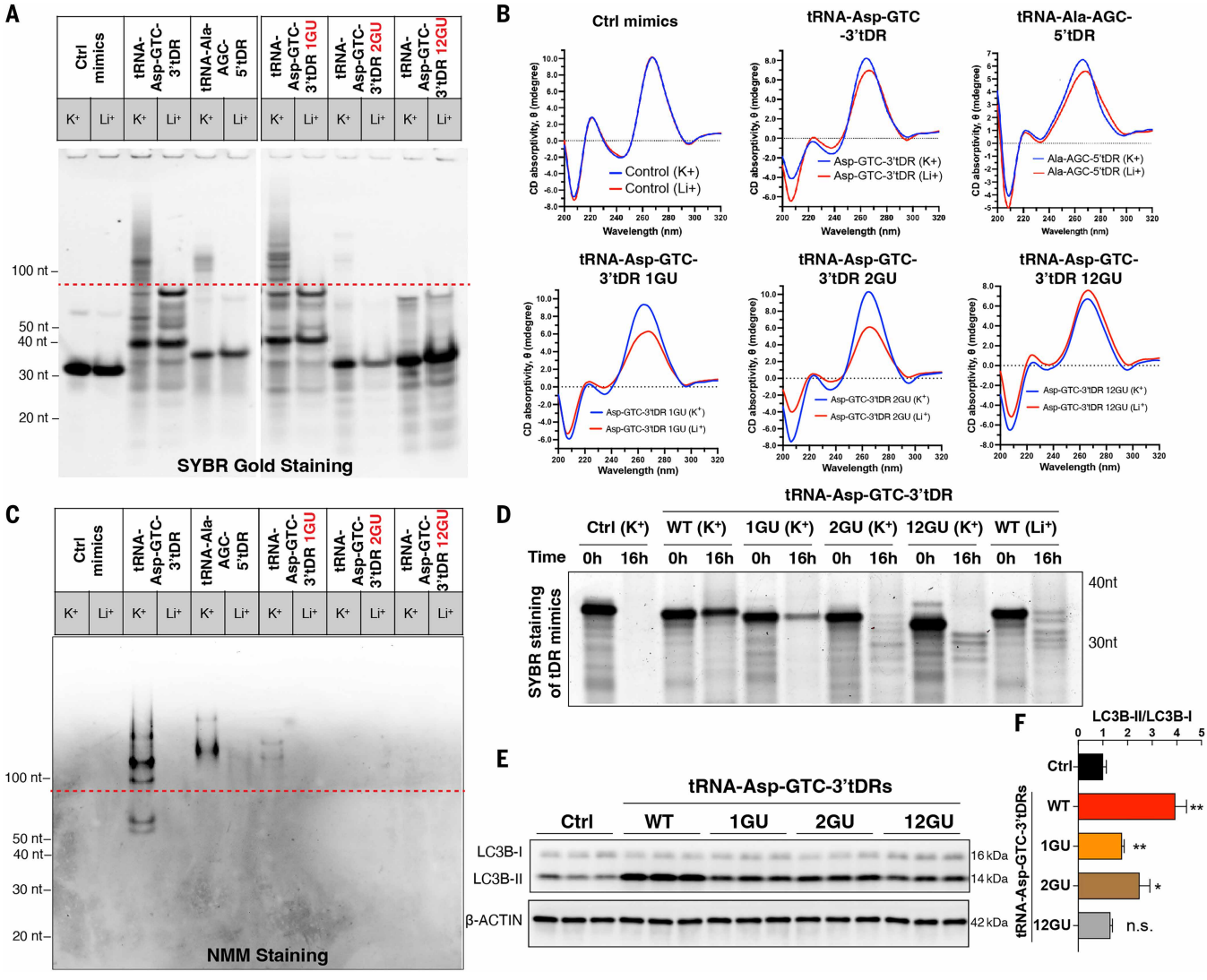


Fig. 6. tRNA-Asp-GTC-3′tDR assembles stable intermolecular G4 structures. (**A**) Electrophoretic mobility shift analysis of intermolecular assemblies under G4 permissive (K^+) or G4-nonpermissive (Li^+) conditions for mimics of Ctrl, tRNA-Asp-GTC-3′tDR, and three 4-guanine motif mutants, with tRNA-Ala-AGC-5′tDR serving as a positive control. SYBR gold was used to stain the gels. 1GU, mutation of the first 4-guanine motif; 2GU, mutation of the second 4-guanine motif; and 12GU, mutation of both 4-guanine motifs. (**B**) CD spectral analysis of monovalent cation responses for mimics of Ctrl, tRNA-Asp-GTC-3′tDR, and three 4-guanine motif mutants, with tRNA-Ala-AGC-5′tDR serving as a positive control. (**C**) NMM staining of intermolecular assemblies under K^+ or Li^+ conditions for mimics of Ctrl, tRNA-Asp-GTC-3′tDR, and three 4-guanine motif mutants, with tRNA-Ala-AGC-5′tDR serving as a positive control. (**D**) SYBR staining showing the stability of assembled tDR mimics after 16 hours of exposure to crude cell lysate. (**E**) Immunoblot analysis of protein levels of LC3B and β-actin in HEK cells transfected with WT or mutants of tRNA-Asp-GTC-3′tDRs. (**F**) Quantification of autophagic protein levels in (E). Data are shown as means ± SEM of at least three independent experiments. Unpaired two-tailed Student's *t* test was used in (F). $*P < 0.05$, $**P < 0.01$.

stability (Fig. 6D). Functionally, 1GU and 2GU mutants showed reduced ability to regulate autophagy compared with wild type (WT), whereas 12GU mutation completely eliminated its regulatory capacity (Fig. 6, E and F, and fig. S7E). Thus, tRNA-Asp-GTC-3′tDR assembles stable intermolecular G4 structures that are critical for its stability and functionality.

tRNA-Asp-GTC-3′tDR sequesters pseudouridine synthase 7 proteins to activate autophagy

The functions of tDRs depend on their protein interactions (*12*–*15*). To identify functional proteins binding to tRNA-Asp-GTC-3′tDR, we transfected biotin-conjugated tRNA-Asp-GTC-3′tDR mimics into HEK cells and performed streptavidin pulldown followed by proteomics analyses. The analysis revealed that pseudouridine synthases (PUS1, PUS4, and PUS7) and DHX36, a G4-specific helicase, are the top four binding proteins of tRNA-Asp-GTC-3′tDR (Fig. 7A). This was confirmed by the probability of protein interaction analysis (table S2) using the SAINTexpress algorithm (*42*) and validated by immunoblotting (Fig. 7B). Analysis of human kidney single-cell RNA-sequencing data (*43*) showed that these genes are abundantly expressed across various kidney cell types (fig. S8A). We then generated stable cell lines expressing two distinct short hairpin RNAs (shRNAs) targeting each candidate. We found that PUS7 knockdown resulting from two separate shRNAs with different knockdown efficiency led to a significant PUS7-dose-dependent elevation in LC3B lipidation, and the additional tRNA-Asp-GTC-3′tDR transfection showed diminished ability to augment LC3B lipidation compared with control conditions (Fig. 7, C and D, and fig. S8B). By contrast, knockdown of PUS1, PUS4, and DHX36 did not appear to notably affect tRNA-Asp-GTC-3′tDR–mediated autophagic flux activation, although we could not exclude a modest effect

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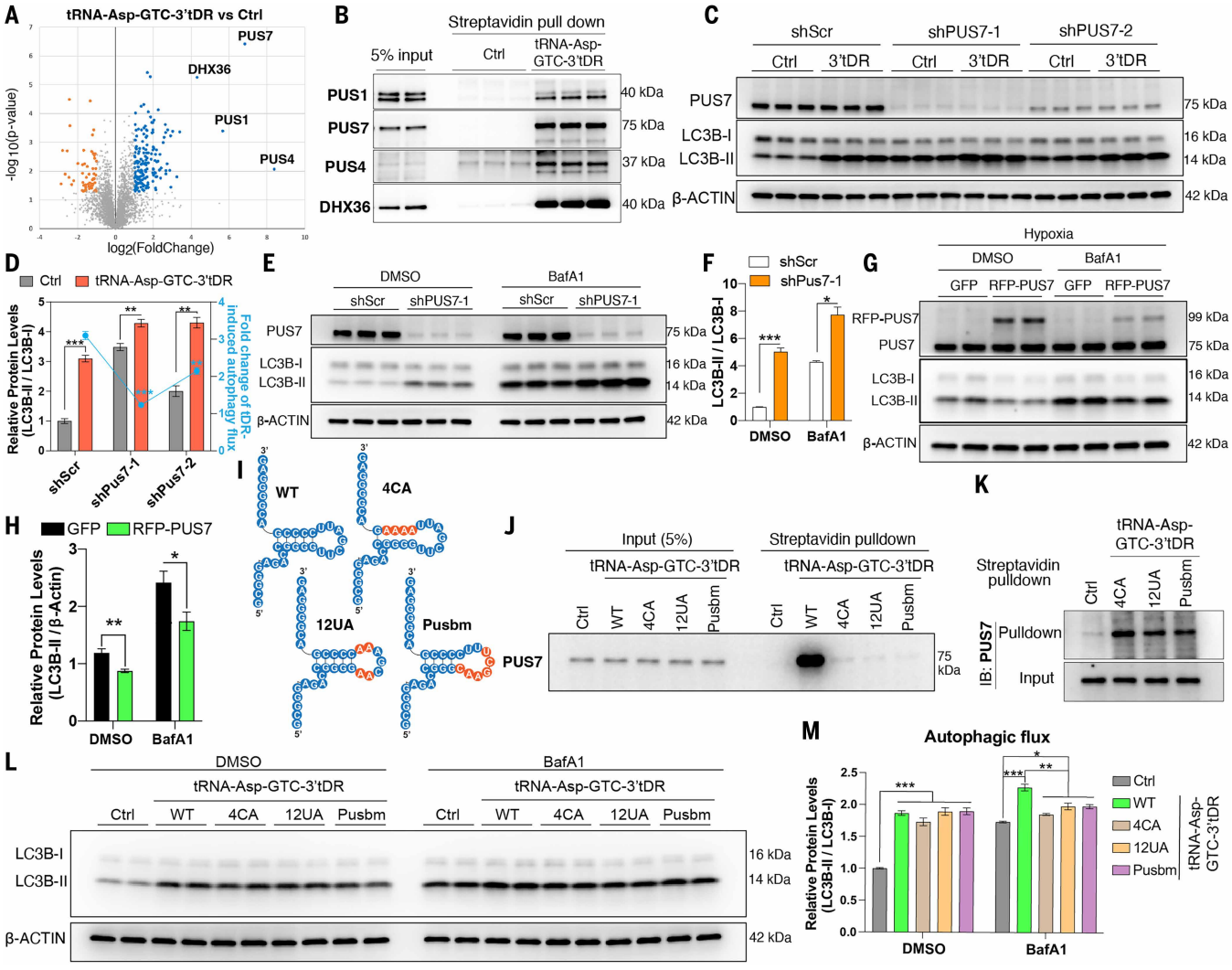


Fig. 7. tRNA-Asp-GTC-3′tDR activates autophagic flux through PUS7 protein sequestration. (**A**) Proteomics analysis of binding proteins of tRNA-Asp-GTC-3′tDR in HEK cells. Significant binding candidates ($P < 0.05$) are highlighted in orange ($\log_2$[fold change] < −1) or blue ($\log_2$[fold change] >1). (**B**) Immunoblot validation of the top four tRNA-Asp-GTC-3′tDR–binding proteins after pull-down of biotin-conjugated Ctrl or tRNA-Asp-GTC-3′tDR mimics from HEK cells using streptavidin beads. (**C**) Immunoblot analysis of PUS7 and LC3B protein levels in scramble shRNA (shScr) and PUS7-silenced (shPUS7-1 and shPUS7-2) HEK cells transfected with Ctrl or tRNA-Asp-GTC-3′tDR (3′tDR) mimics. (**D**) Quantification of protein levels in (C), with the right *y* axis showing the fold change of the LC3B-II/LC3B-I ratio between the tRNA-Asp-GTC-3′tDR and Ctrl groups. (**E**) Immunoblot analysis of PUS7 and LC3B protein levels in Ctrl and PUS7-silenced cells treated with DMSO or BafA1 for 6 hours. (**F**) Quantification of protein levels in (E). (**G**) Immunoblot analysis of PUS7 and LC3B protein levels in PUS7-overexpressing cells treated with DMSO or BafA1 for 6 hours. (**H**) Quantification of protein levels in (G). (**I**) Schematic illustration of three T-arm mutants of tRNA-Asp-GTC-3′tDR. (**J**) Immunoblot analysis comparing PUS7-binding affinity of three T-arm mutations (4CA, 12UA, and Pusbm) versus WT tRNA-Asp-GTC-3′tDR. (**K**) Immunoblot analysis comparing PUS7-binding affinity of 4CA, 12UA, and Pusbm versus Ctrl mimics. (**L**) Immunoblot analysis of LC3B protein levels in HEK cells transfected with different mutants of tRNA-Asp-GTC-3′tDR and treated with DMSO or BafA1 for 6 hours. (**M**) Quantification of protein levels in (L). Data are shown as means ± SEM of at least three independent experiments. Unpaired two-tailed Student's *t* test was used in (D), (F), (H), and (M). $^*P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$.

of PUS4 knockdown (fig. S8, C to H). Thus, PUS7 is the key downstream effector of tRNA-Asp-GTC-3′tDR in modulating autophagy flux.

We next assayed autophagic flux in PUS7-deficient and PUS7-overexpressing cells and found that PUS7 deficiency phenocopied the effects of tRNA-Asp-GTC-3′tDR in enhancing autophagic flux (Fig. 7, E and F, and fig. S9, A and B). Conversely, PUS7 overexpression blocked autophagic flux (Fig. 7, G and H), similar to tRNA-Asp-GTC-3′tDR inhibition. Given that tRNA-Asp-GTC-3′tDR did not affect PUS7 protein levels (Fig. 7C), these results suggested that it enhances autophagic flux by binding and inhibiting PUS7 function. Analysis of cellular fractions from hypoxic HEK cells showed that both PUS7 proteins and endogenous tRNA-Asp-GTC-3′tDR localized to the cytoplasm (fig. S9C), further supporting this hypothesis.

To investigate the essential sequences responsible for PUS7 protein interaction, we first examined the role of G4 structures. Cell-free competitive binding assays revealed that G4-structured tRNA-Asp-GTC-3′tDR markedly increased its binding affinity for PUS7 proteins compared with non–G4-structured tDRs (fig. S10, A and B). The G4-deficient 12GU mutant showed reduced binding affinity for PUS7 (fig. S10C). We further investigated sequence specificity by generating several tRNA-Asp-GTC-3′tDR mutants, including a 4CA mutant (four continuous cytosines in the T-stem substituted with adenines), a 12UA mutant (four potential pseudouridylated uridines in the T-loop replaced with adenines), and a Pusbm mutant [the putative PUS7 RNA–binding GUUCGA motif (*44*) mutated to CAAGCU] (Fig. 7I). Initial characterization demonstrated that all three mutants successfully

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formed stable G4 structures (fig. S10, D to G). However, all three mutants exhibited significantly reduced, but not abolished, binding affinities for PUS7 proteins compared with WT (Fig. 7, J and K), highlighting the crucial role of the T-arm region in PUS7 protein interactions. Functionally, these mutants demonstrated diminished capacities to enhance autophagic flux compared with WT (Fig. 7, L and M), correlating with their reduced (but not abolished) PUS7-binding capacity. Thus, the functionality of tRNA-Asp-GTC-3′tDR is critically dependent on both G4 structure assembly and PUS7 protein interaction.

tRNA-Asp-GTC-3′tDR blocks PUS7-mediated pseudouridylation of histone mRNAs and triggers RNA autophagy

Because PUS7 silencing phenocopies the function of tRNA-Asp-GTC-3′tDR, we hypothesized that PUS7's targeting of other RNA species [including mRNAs, tRNAs, ribosomal RNAs (rRNAs), and small nuclear RNAs (snRNAs) (*45*)] rather than tDR itself may play a crucial role in regulating autophagy flux. We therefore conducted transcriptomic analyses of both long and small RNAs from HEK cells transfected with either control or tRNA-Asp-GTC-3′tDR mimics. Although OTTR-seq, a low-bias small RNA-sequencing method (*8*), revealed no substantial changes in small RNA species, including tRNAs, rRNAs, and snRNAs (fig. S11, A to C), differential expression analysis of long RNAs identified significant changes, with histone transcripts representing the predominant species of the most significantly down-regulated long RNAs (Fig. 8A). Whereas tRNA-Asp-GTC-3′tDR triggered a reduction in histone mRNA levels (Fig. 8B), histone protein levels remained stable (fig. S11D), indicating robust maintenance of histone protein homeostasis. The down-regulation of these histone mRNAs was also observed in HEK

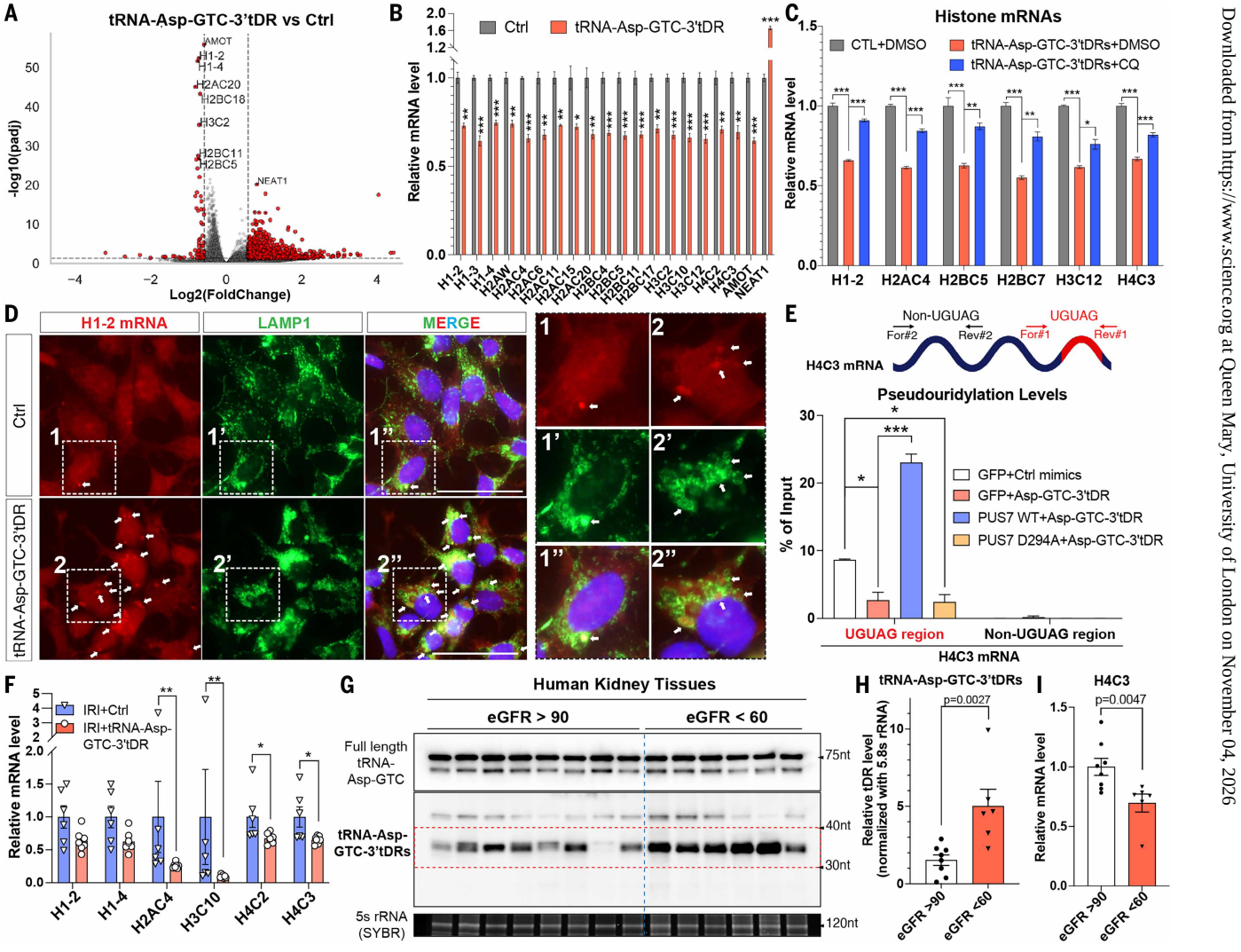


Fig. 8. tRNA-Asp-GTC-3′tDR inhibits PUS7-mediated histone mRNA pseudouridylation and activates RNA autophagy. (**A**) Volcano plot showing the differentially expressed genes in HEK cells transfected with tRNA-Asp-GTC-3′tDR mimics compared with Ctrl mimics. (**B**) Validation of the top 20 significantly regulated genes by tRNA-Asp-GTC-3′tDR treatment. (**C**) Expression analysis of tRNA-Asp-GTC-3′tDR–down-regulated histone mRNAs in HEK cells transfected with tDR mimics and treated with or without CQ for 2 hours. (**D**) RNA FISH assay showing the accumulation of histone H1-2 mRNA (DNA probe; red) in lysosomes (LAMP1 antibody; green) within HEK cells after tRNA-Asp-GTC-3′tDR transfection and CQ treatment. (**E**) Pseudouridylated RNA immunoprecipitation followed by qRT-PCR analysis of H4C3 mRNA pseudouridylation levels at UGUAG and non-UGUAG regions in HEK cells transfected with vectors expressing GFP (Ctrl), PUS7 (WT), or PUS7 D294A (catalytically inactive mutant) and mimics of Ctrl or tRNA-Asp-GTC-3′tDR. (**F**) Expression levels of histone mRNAs in kidney tissues from renal IRI mice treated with Ctrl or tRNA-Asp-GTC-3′tDR mimics. (**G**) Northern blotting showing the levels of tRNA-Asp-GTC-3′tDR in kidney tissues from human patients with high (>90) or low (<60) eGFR. (**H**) Quantification of the levels of tRNA-Asp-GTC-3′tDR in (G). (**I**) Expression analysis of the levels of histone H4C3 mRNA in the kidney tissues of human patients with high (>90) or low (<60) eGFR. Data are shown as means ± SEM of at least three independent experiments. Unpaired two-tailed Student's *t* test was used in (B), (C), and (E). Mann-Whitney test was used in (F), (H), and (I). $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$. Scale bar, 50 μm.

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cells after hypoxia treatment or PUS7 silencing (fig. S11, E and F). PUS7 preferentially modifies uridine within the UNUAR consensus sequence on target RNAs (*46*–*48*). Analysis revealed that 17 of 26 down-regulated histone mRNAs contain one or many UNUAR motifs (table S3), indicating that these histone mRNAs are preferential direct targets of PUS7. Cell-free competitive binding assays confirmed that tRNA-Asp-GTC-3′tDR exhibits stronger binding affinity for PUS7 proteins compared with histone mRNAs (fig. S12, A and B). RNA pseudouridylation plays crucial roles in regulating RNA function and stability (*49*), typically enhancing mRNA stability (*50*). As expected, T-arm mutations with reduced PUS7-binding affinity showed diminished ability to induce histone mRNA degradation compared with WT, and the G4-deficient mutant completely lost its capacity to down-regulate histone mRNAs (fig. S12C). Thus, the sequestration of PUS7 by tRNA-Asp-GTC-3′tDR results in pseudouridylation deficiency in histone mRNAs, leading to their degradation.

Because tRNA-Asp-GTC-3′tDR activates autophagic flux through sequestering PUS7 from histone mRNAs, we conjectured that pseudouridylation-deficient histone mRNAs are directed toward the autophagy pathway for degradation, thereby triggering RNA autophagy. To test this hypothesis, we treated tRNA-Asp-GTC-3′tDR-transfected cells with chloroquine (CQ), an inhibitor that blocks autophagosome-lysosome fusion (*51*). Two hours of CQ treatment substantially slowed tRNA-Asp-GTC-3′tDR–induced histone mRNA degradation (Fig. 8C), indicating an autophagy-associated degradation of these histone mRNAs. We then used an in situ RNA hybridization assay after CQ pretreatment to determine whether the pseudouridylation-deficient histone mRNAs were transported to the autophagosome–lysosome pathway for degradation. We detected a significant accumulation of *H1-2* mRNAs in LAMP1-positive lysosomes in tRNA-Asp-GTC-3′tDR–transfected cells (Fig. 8D). Next, pseudouridylated RNA pulldown using a pseudouridine-specific antibody revealed that the pseudouridylation levels in the UGUAR region of histone mRNAs were significantly diminished in tRNA-Asp-GTC-3′tDR–transfected HEK cells after CQ pretreatment compared with the control (fig. S12D), and the decreased pseudouridylation levels were restored by the overexpression of WT but not dominant-negative PUS7 (Fig. 8E). To investigate whether pseudouridylation-deficient histone mRNAs directly trigger RNA autophagy, we generated in vitro–transcribed *H1-2* mRNAs with either fluorouridine or pseudouridine incorporated in place of uridine to model pseudouridine deficiency or constitutive pseudouridylation, respectively. When transfected to HEK cells, fluorouridine-modified *H1-2* mRNAs significantly enhanced autophagic flux compared with pseudouridine-modified *H1-2* mRNAs (fig. S12, E and F). Thus, tRNA-Asp-GTC-3′tDR/PUS7 interaction results in pseudouridylation deficiency of histone mRNAs, triggering RNA autophagy.

We also investigated this RNA autophagy pathway in our UUO and IRI models. The initial day after UUO surgery showed elevated tRNA-Asp-GTC-3′tDR levels accompanied by a significant reduction in histone mRNA expression (fig. S12G). By day 3 after surgery, histone mRNA levels began to recover as tRNA-Asp-GTC-3′tDR levels moderately decreased (fig. S12G). However, these recoveries were subsequently hindered by the renewed increase in tRNA-Asp-GTC-3′tDR levels after day 5 in the UUO model (fig. S12G). In IRI kidney tissues, tRNA-Asp-GTC-3′tDR administration significantly decreased histone mRNA abundance (Fig. 8F). Thus, tRNA-Asp-GTC-3′tDR negatively regulates histone mRNA levels in mouse kidneys.

Finally, we examined this tDR–histone mRNA axis in human urine and kidney tissues. We analyzed tRNA-Asp-GTC-3′tDR levels in urine samples collected from a cohort of pregnant women with or without preeclampsia. Proteinuria, a hallmark of preeclampsia associated with renal dysfunction (*52*), was present in preeclamptic patients. We observed elevated levels of tRNA-Asp-GTC-3′tDRs in the urine of pregnant women with preeclampsia compared with those without (fig. S13, A and B). Additionally, within patients undergoing total kidney nephrectomy enrolled in the PRECISE cohort (*53*–*55*), patients with early-stage chronic kidney disease (CKD stage 3) with an estimated glomerular filtration rate (eGFR) <60 ml/min/1.73 m^2 were compared with patients with an eGFR >90 ml/min/1.73 m^2 matched for age, gender, and body-mass index (table S4). Kidney tissue RNAs from these groups were then analyzed. The levels of tRNA-Asp-GTC-3′tDR were markedly elevated in patients with CKD stage 3 compared with patients with normal eGFR (Fig. 8, G and H). Moreover, the expression of histone *H4C3* mRNA was significantly reduced in patients with CKD stage 3 compared with patients with normal eGFR (Fig. 8I). Together, these data provide validation for the regulation of tRNA-Asp-GTC-3′tDR in human kidney disease and support an association between tRNA-Asp-GTC-3′tDR and the mRNA autophagy pathway.

Discussion

Although 5′tDRs, encompassing 5′tRNA halves and 5′tRNA fragments, have been detected across tissues and biofluids and extensively studied (*12*, *16*, *56*, *57*), investigations into 3′tDR biology remain scarce. This knowledge gap stems partly from base modifications at the 3′ end of tRNAs, particularly 1-methyladenosine on position 58, which can impede reverse transcription during small RNA sequencing (*5*, *20*). By using ARM-seq, we identified abundant tRNA-Asp-GTC-3′tDR in kidney cells and tissues. tDRs have been functionally linked to translation, ribosome biogenesis, gene silencing, stress granule assembly, and apoptosis (*9*, *11*), and we have also discovered an important role for them in autophagy regulation. We showed that tRNA-Asp-GTC-3′tDR enhanced autophagic flux, whereas its silencing blocked it. The abundance of tRNA-Asp-GTC-3′tDR in kidney cells and tissues, even at basal conditions, aligns with the kidney's high metabolic activity and physiological hypoxia (*58*). Kidneys maintain the highest levels of baseline autophagy activity among organs and rely on autophagy for maintaining their homeostasis, structure, and function (*33*, *59*). Autophagy in kidneys further increases during stress conditions such as exposure to nephrotoxins, sepsis, and ischemia/reperfusion (*60*). Indeed, we found increased tRNA-Asp-GTC-3′tDR levels across multiple murine kidney disease models and in human samples from patients with kidney diseases. Inhibition of tRNA-Asp-GTC-3′tDR in injured kidney tissues using ASOs blocked the autophagy pathway and exacerbated kidney injury, whereas increasing its abundance using synthetic tDR mimics enhanced autophagy activity and was renoprotective. These data support a role for tRNA-Asp-GTC-3′tDR in maintaining basal autophagy in kidney cells and enhancing autophagy as a compensatory response to injury.

Although microRNAs and long noncoding RNAs have been recently implicated in autophagy regulation, primarily by modulating ATG gene expression (*61*), our study unveiled a distinct mechanism for tDRs. First, tRNA-Asp-GTC-3′tDR is a 3′tDR that is dependent on two internal 4-guanine motifs to form G4 structures. Previously, only two 5′tDRs were known to form G4 structures through the 5′-terminal oligo-guanine motif (*12*, *40*). The G4 structures were essential for the stability and functionality of this tDR, and mutations that disrupted the formation of G4 structures had diminished capacity for regulating autophagy. Second, the interaction of tRNA-Asp-GTC-3′tDR with PUS7 was crucial for enhancing autophagic flux. Mutations in the T-arm of tRNA-Asp-GTC-3′tDR significantly reduced both PUS7-binding affinity and the ability to enhance autophagic flux despite their capacity to form G4 structures. Third, tRNA-Asp-GTC-3′tDR induced RNA autophagy. tRNA-Asp-GTC-3′tDR sequestered PUS7 from its target histone transcripts, reducing their pseudouridylation. These pseudouridylation-deficient mRNAs were trafficked into the autophagosome-lysosome pathway for degradation, triggering RNA autophagy. The absence of polyadenylation in histone mRNAs (*62*) may explain their higher sensitivity to pseudouridylation as a mechanism for being targeted by RNA autophagy. The induction of RNA autophagy was associated with enhanced general autophagy flux and appears to be an important adaptive response in kidney cells.

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There are several limitations of our study. First, although ANG-mediated tRNA cleavage typically produces a 5′-OH on 3′tDR and a 3′-cP on 5′tDR and retains extensive modifications from parent tRNAs (*63*), our synthesized tDR mimics conclude with 5′-OH and 3′-OH without the modifications presented on endogenously generated tDRs due to technical challenges and poorly characterized modification patterns and terminal structures. Similarly, the intermolecular G4-forming ability was only determined for synthetic tDR mimics. Future strategies to generate endogenous tDRs may validate these findings definitively. Second, while providing robust evidence for tRNA-Asp-GTC-3′tDR's role in mediating autophagic flux in HEK and other kidney cells, we observed distinct cellular responses between HEK cells and primary kidney cells. Under hypoxia, tDR silencing in HEK cells led to impaired autophagosome maturation, whereas tDR silencing in human kidney cells led to an additional impairment in autophagy initiation. This may reflect inherent differences between HEK cells and adult kidney cells (*64*–*66*). Third, our investigation of the tDR/PUS7 axis focused primarily on histone mRNAs, leaving the effects on other down-regulated transcripts unexplored. We also observed up-regulation of other RNA species, including mRNAs, long noncoding RNAs, rRNAs, and snRNAs, although the underlying mechanisms, whether through pseudouridylation or other pathways, require further investigation. Finally, the definitive mechanistic connection between RNA autophagy and general autophagy remains to be elucidated.

Materials and methods

Cell culture

HEK293 cells were purchased from ATCC (CRL-1573) and cultured in D10 medium, which consisted of DMEM with high glucose and pyruvate (Thermo Fisher, catalog no. 11995073) supplemented with 10% fetal bovine serum (FBS) (Thermo Fisher, catalog no. 16000044) and 1% penicillin/streptomycin (Thermo Fisher, catalog no. 15140122). For the hypoxia treatment, HEK cells were fed with complete D10 medium and maintained in hypoxia chamber with 0.1% oxygen for 24 hours. Human primary renal proximal tubule epithelial cells were purchased from Lonza (Lonza, CC-2553) and cultured in the renal epithelial cell growth medium (Lonza, CC-3190). Human primary kidney fibroblasts were purchased from Cell Biologics (Cell Biologics, H-6016) and cultured in the complete fibroblast medium (Cell Biologics, M2267). Human primary glomerular microvascular endothelial cells were purchased from Cell Systems (Cell Systems, ACBRI 128 V) and cultured in the complete classic medium with serum and Culture-Boost (Cell Systems, 4Z0-500). Primary human renal glomerular epithelial cells were purchased from Cell Applications (Cell Applications, 942-05n) and cultured in the renal epithelial cell growth medium (Lonza, CC-3190). Primary human glomerular mesangial cells were purchased from Cell Systems (Cell Systems, ACBRI 127 V) and maintained in the complete classic medium with serum and Culture-Boost (Cell Systems, 4Z0-500).

Human immortalized podocyte cell line (*67*) was a gift kindly provided by Dr. Moin A. Saleem from the University of Bristol and cultured in R10 medium, which consists of RPMI-1640 basal medium with GlutaMAX (Thermo Fisher, catalog no. 61870127) supplemented with 10% FBS (Thermo Fisher, catalog no. 16000044), 1% Insulin-Transferrin-Selenium (Thermo Fisher, catalog no. 41400045), and 1% Penicillin/Streptomycin (Thermo Fisher, catalog no. 15140122). Podocytes were first amplified at the 33°C CO_2 incubator and matured by culturing them at a 37°C CO_2 incubator for 2 weeks before experiments.

To block the autophagic flux in these cells, 20 nM BafA1 or 50 μM CQ was added to the cell culture medium for 6 hours or 2 hours, respectively, before collecting the cells for RNA or protein analysis.

In vitro nucleic acid oligo transfection

For the transfection of RNA and DNA oligos, 50 nM mimics (synthesized by Integrated DNA Technologies), 50 nM tDR biogenesis reporter (synthesized by Dharmacon), or 100 nM ASOs (synthesized by Qiagen) were transfected into the cells using siPORT NeoFx (Thermo Fisher, catalog no. AM4511) or Lipofectamine RNAiMax (Thermo Fisher, catalog no. 13778150) by following the manufacturer's instructions, unless otherwise specified. tDR mimics synthesized at Integrated DNA Technologies at default mode end with 5′-OH and 3′-OH and no additional modification was added.

The concentration of 50 nM for tDR mimics was determined by comparing the levels of transfected tRNA-Asp-GTC-3′tDR mimics at different concentrations with its physiological levels in hypoxic HEK cells. The analysis indicated that the physiological level of tRNA-Asp-GTC-3′tDR in hypoxic HEK cells corresponds to ~1 nM (fig. S14). Considering that lipid nanoparticle-mediated RNA delivery has limited endosomal escape efficiency of only 1~2%, with most RNA/particle complexes remaining trapped in the endosomal vesicles before eventual lysosomal degradation (*68*, *69*), our transfection protocol using 50 nM tDR mimics with Lipofectamine translates to an effective cytoplasmic delivery of 1 nM (2%) functional tDRs,

Generation of stable transgenic cell lines

For gene knockdown, shRNAs targeting the mRNA of PUS1, PUS4, PUS7, or DHX36 were designed and cloned into the pLKO.1-TRC cloning lentiviral vector, a gift from David Root (Addgene plasmid no. 10878). A scramble shRNA was also designed as a control. The shRNA sequences are listed in table S5. For building autophagy reporter cell lines, GFP-LC3B and RFP-GFP-LC3B DNA fragments were synthesized by GenScript and subcloned into pLV-EF1a-IRES-puro lentiviral vector, a gift from Tobias Meyer (Addgene plasmid no. 85132). For PUS7 overexpression, GFP, RFP-PUS7, and RFP-PUS7 D294A DNA fragment were synthesized by GenScript and subcloned into the pLV-EF1a-IRES-puro lentiviral vector. For lentivirus packaging, lentiviral vector DNA (10 μg) was transfected into HEK 293T cells (one of 100 mm dish) together with the packaging plasmids, ps-PAX2 (7.5 μg) and pMD2.G (5 μg), using the Lipofectamine 3000 (Thermo Fisher, catalog no. L3000015) or PEI-max (Polysciences, catalog no. 24765) as previously described (*70*). The supernatant containing the lentiviruses was concentrated using the Lenti-X concentrator (Takara, catalog no. 631232) and then added to HEK cells supplied with 8 μg/ml polybrene. Stable transgenic cell lines were selected by 10 μg/ml puromycin for 3 to 6 days. ANG-knockout HEK cells were generated using paired gRNAs, as mentioned previously (*20*).

Animal studies

WT C57BL/6 mice were purchased from Jackson Laboratory (strain 000664) and maintained under pathogen-free conditions. All procedures conformed to the animal welfare regulations of the Massachusetts General Hospital or Brigham and Women's Hospital Sub-Committees on Animal Research Care and under the guidelines on the use and care of laboratory animals for biomedical research published by the National Institutes of Health (No. 85-23, revised 1996). All mice used for experiments were within 8 to 14 weeks of age. For examining tDR abundance in different mouse models, 3 to 4 mice were used per group. For tDR GOF/LOF studies in mouse models, 4 to 7 mice were used per sham group, and 6 to 10 mice were used per surgery group. The detailed number of mice used per experiment is indicated in the figures or figure legends.

To perform uninephrectomy, male 10 to 12 weeks of age C57BL/6 mice were anesthetized with isofluorane by inhalation. A 1-cm lumbar incision was performed, and one of the kidneys was exposed after separating the muscle and fascia. The renal vein and ureter from the kidney were occluded together by two ligature knots of 6.0 polyglycolic acid absorbable suture. The kidney was then removed by cutting the renal vessels between the two knots. The muscle layer was then closed using absorbable surgical sutures vicryl and the outer skin was closed with nylon sutures. For the sham surgery, perform the same surgery

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procedures, including exposure of the kidney, dissection of tissue, and wound closure, but without the removal of the kidney.

To induce a protein-overloaded model, male 10 to 12 weeks of age C57BL/6 mice were intraperitoneally injected with endotoxin-free BSA (Sigma-Aldrich, catalog no. A9430) for 7 consecutive days with increasing doses (2, 4, 6, 8, 10, 10, 10 mg/g body weight). PBS was used as the vehicle to dissolve the BSA.

To induce the UUO model, 10- to 12-week-old male C57BL/6 mice were anesthetized with pentobarbital sodium (50 mg/kg, i.p.), 1 cm of incision was performed on the left back of the mice and the ureter of the left kidney was ligated with 2 silk threads 3.0. After ureteral obstruction, the muscle layer was closed using absorbable surgical sutures vicryl, and the outer skin was stapled using sterile staples or closed with nylon sutures. Mouse kidneys were collected on day 6 post-UUO surgery. For the sham surgery, perform the same surgery procedures, including exposure of the kidney, dissection of tissue, and wound closure, but without ureteral obstruction.

To induce the renal IRI model, 10- to 12-week-old male C57BL/6 mice were anesthetized with pentobarbital sodium (50 mg/kg, i.p.). Uninephrectomy was first performed to remove the right kidney. A week later, mice were anesthetized with isoflurane by inhalation. A 1 cm of incision was performed on the left back of the mice, and the pedicle of the left kidney was clamped with nontraumatic microaneurysm clamps (Roboz) for 18 min and then released. After reperfusion, the muscle layer was closed using absorbable surgical sutures vicryl, and the outer skin was stapled using sterile staples or closed with nylon sutures. Mouse kidneys were collected on day 6 post-IRI surgery. For sham surgery, perform the same surgery procedures, including exposure of the kidney, dissection of tissue, and wound closure, but without uninephrectomy or pedicle clamping.

In vivo ASO administration

ASOs with fully phosphorothioate-modified backbones were synthesized by Qiagen. The sequences of ASOs are the same as those used in vitro and detailed in table S5. Modified ASO targeting tRNA-Asp-GTC-3′tDR or negative control was injected (30 mg/kg body weight, i.v.) into mice undergoing sham or UUO surgery beginning one day before surgery and continuing every other day until day 6 post-UUO surgery. For baseline inhibition, modified ASO targeting tRNA-Asp-GTC-3′tDR or negative control was injected subcutaneously (30 mg/kg body weight) into mice without surgery once a week for 4 weeks. Modified ASO targeting tRNA-Asp-GTC-3′tDR or negative control ASO were injected (30 mg/kg body weight, i.v.) into mice beginning five days before IRI surgery and continuing every other day until day 4 post-UUO surgery.

In vivo delivery of tDR mimics

RNA mimics of tRNA-Asp-GTC-3′tDR and control sequences were synthesized by Integrated DNA Technologies using standard parameters, yielding molecules with 5′-OH and 3′-OH termini without additional modifications.

For DharmaFECT-mediated in vivo delivery, tDR mimics (1.5 mg/kg body weight) were combined with DharmaFECT1 transfection reagent (Dharmacon, catalog no. T-2001-03) in RNase-free water, incubated for 15 min at room temperature, and subsequently resuspended in 100 μl RNase-free water for intravenous injection.

For PEI-mediated in vivo delivery, transfections were performed following the in vivo jetPEI protocol (Polyplus, catalog no. 101000030), maintaining a PEI nitrogen to RNA phosphate molar ratio of 8. For a 30 g mouse receiving 3 mg/kg body weight: tDR mimics (9 μl of 10 mg/ml stock in TE buffer) were diluted in 91 μl of RNase-free 5% glucose solution in one tube, and 14.4 μl of in vivo jetPEI or 150 mM PEI-MAX (nitrogen molar concentration; Polysciences, catalog no. 24765) was diluted in 85.6 μl of RNase-free 5% glucose solution in a separate tube. The contents of both tubes were mixed thoroughly and incubated at room temperature for 15 min. The resulting 200 μl mixture was administered through intravenous injection. For the 1.5 mg/kg body weight dose, a 100 μl mixture was used. Injection volumes were proportionally adjusted according to individual mouse weights. In the renal IRI model, tDR/PEI mixtures were administered (3 mg/kg body weight, i.v.) to sham or IRI mice beginning five days before IRI surgery and continuing every other day until day 6 post-IRI surgery.

Patient sample collection

Kidney tissue samples were obtained from patients undergoing total nephrectomy enrolled in the PRECISE cohort at the University of Michigan (institutional review board numbers: HUM00165536 and HUM00052918). Patient information is described in table S4. Kidney tissue was placed in preservatives shortly after extraction of the kidney from patients to minimize ischemic time. Tissue samples were stored in RNAlater (Thermo Fisher, catalog no. AM7021) and frozen at −80°C until use. Clinical data were abstracted using the honest broker system as described (*53*) or from the participant's electronic medical record.

Urine samples were collected from patients with and without preeclampsia with written informed consent under IRB protocol 181917, approved by the Human Research Protections program at the University of California, San Diego (UC San Diego). Proteinuria was defined as ≥1+ on urine dip or urinalysis on 2 occasions, ≥0.3 on P/C ratio, or ≥300 mg total protein on 24-hour urine collection. Clean-catch urine samples were collected into sterile containers, centrifuged at 2000 X g for 10 min at room temperature to pellet cells and cellular debris, and the supernatant was removed, aliquoted, and stored at −80°C within 2 hours of collection until further processing. The miRCURY Exosome Cell/Urine/CSF Kit (Qiagen, catalog no. 76743) was used to precipitate extracellular vesicles from 4 ml of urine using the manufacturer's protocol, followed by RNA extraction using Qiazol and miRNeasy Micro Kit spin columns using the manufacturer's protocol (Qiagen, catalog no. 217084). Small RNA libraries were prepared for sequencing as previously described (*71*), using the NEB Next Small RNA Library Prep Set for Illumina (New England Biosciences, catalog no. E7330L) according to the manufacturer's protocol with the following modifications. All reactions were conducted at 1/5th the volume with 1.2 μl of RNA input, adapters were diluted 1:6 of the supplied concentration, and 18 PCR cycles were conducted. Small RNA libraries were cleaned using the Zymo DNA Clean and Concentrate kit (Zymo Research, catalog no. D4103), pooled, and size-selected using the Pippin Prep HT with a 125-150bp size limit to remove adapter dimers. Size-selected libraries were sequenced on the Illumina NovaSeq 6000 as 75bp single-end reads by the Institute for Genomic Medicine at UC San Diego (funded by National Institutes of Health SIG grant S10OD026929). Small RNA-Seq data were processed using the Genboree exceRpt Small RNA-Seq Pipeline (*72*). Briefly, FASTQ files underwent adapter trimming and mapping to the endogenous human genome (hg38) using STAR v2.4.2a, with a minimum insert length set to 15 nucleotides and no mismatches permitted on the Genboree workbench, and tRNA counts were extracted.

Quantitative reverse transcription polymerase chain reaction

Quantitative reverse transcription polymerase chain reaction (qRT-PCR) was performed as previously described (*20*). Briefly, cells or tissues were immediately lysed by adding TRIzol (Thermo Fisher, catalog no. A33251) after collection, and total RNAs were isolated by following the manual. One microgram of total RNA was reverse transcribed to cDNA using the High-Capacity cDNA Reverse Transcription Kit (Thermo Fisher, catalog no. 4368813) or The SuperScript III First-Strand Synthesis System (Thermo Fisher, catalog no. 18080051). qPCR was then performed using a QuantStudio 6 Flex Real-Time PCR Systems (Thermo Fisher) with SsoAdvanced Universal SYBR Green Supermix (BioRad, catalog no. 1725275). The sequences of qPCR primers used are listed in table S5.

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Northern blotting

Northern blotting was performed as previously reported (*20*). Briefly, denatured 0.1 ~5 μg total RNAs were separated by 15% Criterion TBE-Urea PreCast Gels (Bio-Rad, catalog no. 3450092 or 3450093). The gels were stained with SYBR-Gold (Thermo Fisher, catalog no. S11494) and transferred onto positively charged Nylon membrane (Sigma-Aldrich, catalog no. 11209299001) using Trans-Blot Turbo Transfer System (Bio-Rad). The membranes were then crosslinked with EDC (Sigma-Aldrich, catalog no. E7750) at 60°C for 1~2 hours and prehybridized with ULTRAhyb Ultrasensitive Hybridization Buffer (Thermo Fisher, catalog no. AM8669). 50 pmol/ml Biotin-labeled LNA-modified DNA probes (synthesized by Qiagen) were used for hybridization at 37°C overnight. After washing sequentially with low-stringent buffer, high-stringent buffer, and 1× SSC, the blots were then processed and developed using Chemiluminescent Nucleic Acid Detection Module Kit (ThermoFisher, catalog no. 89880). For all Northern blots, we used a single probe capable of detecting both full-length tRNA-Asp-GTC and its corresponding tDRs within one intact plot. The complete Northern blot images from all figures are provided in figs. S15 and S16. The sequences for probes are provided in table S5.

Immunoblotting

Immunoblotting was performed as previously described (*20*). Briefly, cell pellets were resuspended in RIPA buffer (Thermo Fisher, catalog no. 89900) containing Protease and Phosphatase Inhibitor Cocktail (Thermo Fisher, catalog no. 78441), and PMSF, and lysed by sonication. Tissue samples were cut into small pieces and homogenized in RIPA buffer containing Protease and Phosphatase Inhibitor Cocktail and PMSF, followed by sonication. The protein concentration of clarified cell or tissue lysate was then measured using the BCA protein assay kit (Thermo Fisher, catalog no. 23227). 5 ~ 20 μg of total protein per lane was separated by SDS-PAGE gel and transferred to PVDF membrane. Then, 5% BSA in TBST was used for blocking and primary antibody incubation. HRP-conjugated secondary antibodies were then applied, and the blots were developed using SuperSignal West Femto Maximum Sensitivity Chemiluminescent Substrate (Thermo Fisher, catalog no. 34094). The antibodies used are listed in table S6.

Live imaging

For live imaging of lysosomal activity (using Lysosomal Intracellular Activity Assay Kit, Abcam, ab234622), autophagy (GFP-LC3B reporter cells), and autophagic flux (RFP-GFP-LC3B reporter cells), cells were incubated with Hoechst 33342 (Thermo Fisher, catalog no. H3570) for 5-15 min and fed with fresh medium, within which the basal medium was replaced with FluoroBrite DMEM (Thermo Fisher, catalog no. A1896701). Images were captured using the Leica SP8 confocal laser scanning microscope system.

Fluorescence in situ hybridization

For the RNA fluorescence in situ hybridization (FISH) assay, cells were pretreated with CQ for 2 hours, followed by fixation in 3.7% formaldehyde (in PBS) and permeabilization in ice-cold methanol at 4°C for 20 min. After washing, cells were incubated with LAMP1 antibody (ThermoFisher, catalog no. MA5-29385) diluted in 1× PBST at 4°C overnight, followed by the incubation of fluorescent secondary antibody. Immunostained cells were fixed again in 3.7% formaldehyde (in PBS) and washed with 1× PBST twice and 1× Stellaris RNA FISH Wash Buffer A (LGC Biosearch Technologies, catalog no. SMF-WA1-60) once. Cells were then incubated with 1× Stellaris RNA FISH Hybridization Buffer (LGC Biosearch Technologies, catalog no. SMF-HB1-10) containing 125 nM Cy3-labeled histone H1-2 DNA probe (synthesized by GenScript) at 37°C for 4-16 hours, followed by washing with 1× Stellaris RNA FISH Wash Buffer A once and 1× Stellaris RNA FISH Wash Buffer B (LGC Biosearch Technologies, catalog no. SMF-WB1-20) once. The cells were then mounted by ProLong Glass Antifade Mountant with NucBlue Stain solution and subjected to Leica SP8 for imaging.

Mouse kidney tissue histology

Kidney tissues of mouse models with the injection of ASOs or tDR mimics were perfused with cold PBS before harvesting. Half of the kidneys were fixed in formalin, dehydrated in 70% EtOH, and embedded in paraffin. Paraffin-embedded tissues were cut into 5- to 7- μm sections and stained with the Hematoxylin and Eosin Staining Kit (Abcam, catalog no. ab245880) or the Picrosirius Red staining kit (Polysciences, catalog no. 09400; or Abcam, catalog no. AB150681). Another half of the kidneys were fixed in 4% PFA, dehydrated in 20% Sucrose, and embedded in optimal cutting temperature (OCT). OCT-embedded tissues were cut into 5- to 10- μm sections and subjected to immunofluorescent staining. Briefly, the frozen sections were permeabilized in 1× PBS containing Triton X-100 (0.1%) for 5 to 15 min and labeled with primary antibodies, followed by the incubation of secondary fluorescent antibodies. ProLong Glass Antifade Mountant with NucBlue Stain solution (Thermo Fisher, catalog no. P36981) was then used for nuclei staining and mounting.

Transmission electron microscopy

Transmission electron microscopy (TEM) was performed in the PMB Microscopy Core at Massachusetts General Hospital. Briefly, after perfusion/immersion fixation with periodate-lysine-paraformaldehyde reagent (4% paraformaldehyde, 75 mM lysine, 10 mM periodate fixative in 0.15 M sucrose, 37.5 mM sodium phosphate), small pieces of kidney tissues with the injection of ASOs were immersed in 2.0% glutaraldehyde in 0.1 M sodium cacodylate buffer, pH 7.4. They were postfixed for 1 hour in 2% osmium tetroxide, dehydrated through a graded series of ethanol solutions up to 100%, and embedded in Epon (Electron Microscopy Sciences, Fort Washington, PA). Thin (~70 nm) sections were collected onto copper slot grids and stained with uranyl acetate and Reynold's lead citrate before examination. Sections were examined in a JEM-1011 transmission electron microscope (JEOL, Tokyo, Japan) at 80 kV.

Electrophoretic gel shift assay

tDR mimics were purchased by Integrated DNA Technologies and dissolved in RNase-free water to prepare 100 μM stock solutions. tDR mimics were further diluted to 10 μM in the indicated salt solution containing either 150 mM KCl or 150 mM LiCl (in 10 mM Tris-HCl pH 7.4, 0.1 mM EDTA buffer). Samples were heated to 95°C for 5 min and then slowly cooled to room temperature using a Touchdown program in a Bio-Rad PCR cycler to facilitate proper folding. For native gel analysis, 20 pmol of folded RNA samples were electrophoresed on a 20% nondenaturing gel. Gels were post-stained in a solution with 1× SYBR Gold (Thermo Fisher, catalog no. S11494) or 50 μM NMM (Santa Cruz Biotechnology, catalog no. sc-396879A) in 0.5× TBE for 10 min and visualized using a 305 nm UV transilluminator.

Circular dichroism spectroscopy

CD spectra were acquired using a JASCO J815 spectropolarimeter. tDR mimics were prepared at 10 μM in the buffer containing either 150 mM KCl or LiCl in TE buffer, as described above. Measurements were performed in quartz cuvettes (1 mm path length) using 200 μl sample volumes. Spectra were recorded from 200 to 320 nm at 20°C, with each spectrum representing an average of three scans. G4 formation was indicated by enhanced peak intensity at 260-265 nm and a trough at 240 nm in K^+ conditions, with reduced peak intensity observed in Li^+ conditions.

Test tube equilibrium complex concentration analysis

To evaluate the targeting specificity of ASO designed for targeting tRNA-Asp-GTC-3′tDR, we developed a computational tube reaction

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model incorporating four components at equal concentrations: ASO targeting tRNA-Asp-GTC-3′tDR, tRNA-Asp-GTC-3′tDR, tRNA-Asp-GTC-5′tDR, and full-length tRNA-Asp-GTC. The secondary structure calculations for this four-component interaction network were carried out using the NUPACK Python package (version 4.0.1.7). The rna06 parameter was used. The tube analysis mode was used, with each component set to a universal initial concentration of 1 μM under physical and chemical conditions of 37°C and 0.15M NaCl to mimic the cellular environment. For the analysis, the equilibrium complex concentrations were calculated using the tube analysis with the maximum complex size set to 2.

Total RNA-seq

Total RNA-seq was performed as previously described (*73*). Briefly, total RNAs from HEK cells transfected with control mimics or tRNA-Asp-GTC-3′tDR mimics were isolated with TRIzol (Thermo Fisher, catalog no. A33251) by following the manual. cDNA libraries were constructed using the SMARTer Stranded Total RNA-Seq Kit v2 Pico Input Mammalian (Takara Bio, San Jose, CA, USA) and sequenced using the Illumina NextSeq 2000 platform using a resolution at 100 base pairs with paired-end reads. Reads were trimmed to remove adapter sequences using Cutadapt (v2.10) and aligned to the Gencode GRCh38.p13 genome using STAR (v2.7.8a). Gencode v38 gene annotations were provided to STAR to improve the accuracy of mapping. Quality control on both raw reads and adapter-trimmed reads was performed using FastQC (v0.11.9) (www.bioinformatics.babraham.ac.uk/projects/fastqc). featureCounts (v2.0.2) was used to count the number of mapped reads to each gene. Heatmap3 was used for cluster analysis and visualization. Significantly differential expressed genes with absolute fold change ≥1.5 and FDR-adjusted $P < 0.05$ were detected by DESeq2 (v1.30.1).

Ordered two-template relay small RNA sequencing

For ordered two-template relay small RNA sequencing (OTTR-seq), from HEK cells transfected with control or tRNA-Asp-GTC-3′tDR mimics, total RNAs were purified from TRIzol (ThermoFisher, catalog no. 15596026) treated cells and then incubated with T4 PNK (NEB, catalog no. M0201L) to resolve 2′-3′ cyclic phosphate ends, and then deacylated by bringing the reaction to pH 9.0 for 30 min at 45°C using 100 mM Na2B4O7. Treated RNA was recovered using the RNA clean and concentrate kit (Zymo Research, catalog no. R1013). OTTR-seq libraries were prepared according to previously described methods (*8*) with modifications. Briefly, 40 ng of the processed total RNA was used as input. RNA was labeled for 2 hours at 30°C using only dideoxyATP (ddATP). Similarly, cDNA synthesis was conducted using only the RNA-DNA primer duplex with +1T overhang. cDNA was recovered using the MinElute reaction cleanup kit (Qiagen, catalog no. 28204), then size selected on a 1×Tris-borate-EDTA (TBE, 89 mM Tris, 89 mM boric acid, 2 mM EDTA, pH 8.3) buffer, 8M urea, and 9% acrylamide gel to remove adapter dimers. cDNA was recovered by diffusion at 70°C for 1 hour with shaking in a solution of 10 mM Tris-HCl pH 8, 300 mM NaCl, 1mM EDTA, and 0.25% SDS, followed by precipitation in 66% ethanol with 0.6 M ammonium acetate and 1 μg/ml Glycoblue (ThermoFisher, catalog no. AM9515). cDNA was amplified with Q5 polymerase (NEB, catalog no. M0493L) for 13 cycles using the Illumina NebNext sRNA index adapters (NEB, catalog no. E7600S) then size selected on a 1× TBE, 6% acrylamide gel to remove remaining adapter dimer. Libraries were recovered from the gel by diffusion at 70°C for 1 hour with shaking in a solution of 10 mM Tris-HCl pH 8, 300 mM NaCl, and 1 mM EDTA, followed by precipitation in 70% ethanol with 80 mM sodium acetate and 1 μg/nl Glycoblue (ThermoFisher, catalog no. AM9515). The recovered libraries were quantified using the qubit dsDNA High Sensitivity kit (Invitrogen, catalog no. Q32854), Agilent High Sensitivity DNA bioanalyzer kit (Agilent, catalog no. 5067-4626), and finally by qPCR. qPCR was conducted using PrimeTime Gene Expression Master Mix (IDT, catalog no. 1055772) according to a previously published protocol40. Libraries were sequenced on the NextSeq 1000/2000 platform to generate 100 bp single-end reads. Raw fastq files were trimmed using Cutadapt with options -j 8 -m 15 -a GATCGGAAGAGCACACGTC. Reads were further processed to remove an 8NT UMI using umi_tools extract with options–extract-method=string–bc-pattern=NNNNNNN. Trimmed reads were processed through the tRNA Analysis of eXpression pipeline (tRAX)41, using a custom reference database generated for the HG38 assembly which includes sequence and annotations for the scrambled Asp-GTC 3′ tDR sequence. Resulting BAM files were split into reads above or below 60bp in length, and those above 60bp were processed through the tRAX pipeline again (using the –lazyremap option) to analyze the expression of mature tRNA transcripts.

In vitro transcription

Human histone H1-2 cDNA, together with an exogenous tag sequence, was synthesized by GenScript and cloned into a pcDNA3.1 vector under a T7 promoter. The linear template T7/H1-2 DNA was then amplified using PrimerSTAR GXL DNA polymerase (Takara Bio, catalog no. R050B) and cleaned up using Wizard SV Gel and PCR Clean-Up System (Promega, catalog no. A9282). H1-2 mRNAs with different uridine modifications were in vitro transcribed using the MEGAscript T7 Transcription Kit (ThermoFisher, catalog no. AM1334) with some modifications. Uridine-5′-triphosphate was replaced with Pseudouridine-5′-triphosphate (Advnt Biotechnologies, catalog no. 13944) or Fluorouridine-5′-triphosphate (Sierra Bioresearch, catalog no. FUTP) to generate Pseudouridine-modified or Fluorouridine-modified H1-2 mRNAs, respectively. The reactions were carried out at 37°C for 16 hours, and the template DNA was removed by TURBO DNase treatment. The H1-2 mRNAs were cleaned up using the Qiagen RNeasy Midi Kit (Qiagen, catalog no. 75144).

tDR pull-down and proteomics

To identify the binding proteins of tRNA-Asp-GTC-3′tDR, 50 nM Biotin-conjugated tRNA-Asp-GTC-3′tDR or control mimics were transfected to HEK cells using Lipofectamine RNAiMAX. 24 hours after transfection, the cells were pelleted after washing and proceeded with RNA pull-down using part of the Pierce Magnetic RNA-Protein Pull-Down Kit (Thermo Fisher, catalog no. 20164). Binding proteins were directly eluted using 1× Laemmli sample buffer (BioRad, catalog no. 161-0747) and subjected to in-solution protein digestion using the SP3 protocol (*74*, *75*). The resulting tryptic peptides were directly loaded on a 50 cm C18 EasySpray column and separated by acidic reverse phase chromatography over a 75 min gradient. Data acquisition was carried out on a Thermo Orbitrap Eclipse mass spectrometer equipped with a FAIMS Pro device using the following settings: FAIMS CV at −40 V, −60 V, and −80 V, MS1 resolution of 120 K, full scan range of 375 to 1575 m/z, precursor isolation window of 1.6, MS2 scans in the iontrap from HCD fragmentation of precursor ions with 32% NCE and dynamic exclusion setting of 60 s. Data were acquired in top-speed mode with a cycle time of 1 s per FAIMS CV. Data were searched in Proteome Discoverer version 2.4 running Mascot version 2.6 against a UniProtKB/Swiss-Prot human protein sequence database (downloaded May 2020). Search parameters included: Cys carbamidomethylation as a fixed and Met oxidation as a dynamic modification; precursor mass tolerance of 10 ppm and a fragment mass tolerance of 0.6 Da. PSM and peptide identifications were filtered to 1% FDR using the Percolator node in Proteome Discoverer version 2.4. The probability of protein interaction to tRNA-Asp-GTC-3′tDR was calculated using SAINTexpress (*48*) and compared against the control group. As recommended by SAINTexpress, high-confidence interactors were thresholded at the average probability of interaction (AvgP) > 0.7.

Competitive PUS7-binding assay

To compare PUS7 binding affinities between histone mRNAs and tDR variants relative to WT tRNA-Asp-GTC-3′tDR, biotin-conjugated

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tRNA-Asp-GTC-3′tDR mimics (50 nM) were transfected into HEK cells. The biotin-tDR/PUS7 complexes were isolated using the Pierce Magnetic RNA-Protein Pull-Down Kit (Thermo Fisher, catalog no. 20164). After washing three times with 1× Washing Buffer, the bead-immobilized biotin-tDR/PUS7 complexes were incubated with either G4-structured tRNA-Asp-GTC-3′tDR mimics, non–G4-structured tRNA-Asp-GTC-3′tDR mimics, control mimics, or varying concentrations of in vitro transcribed histone H1-2 mRNAs. The incubation was performed at 37°C for 30 min in 1× Protein-RNA binding buffer, followed by three additional washes with 1× Washing Buffer. Both the flow-through fraction (containing displaced PUS7) and the bead-bound fraction (containing remaining PUS7) were analyzed by immunoblotting.

Pseudouridylated RNA immunoprecipitation

HEK cells transfected with control or tRNA-Asp-GTC-3′tDR mimics were treated with 50 μM CQ for 2 hours before RNA collection, and the total RNAs were isolated using Qiagen RNeasy Midi Kit (Qiagen, catalog no. 75144). Total RNAs were fragmented using RNA fragmentation Reagent (ThermoFisher, catalog no. AM8740) at 95°C for 5 min, immediately followed by the addition of EDTA and rapid cooling on ice. Fragmented total RNAs were then cleaned up using the RNA Clean and Concentrator kit (Zymo Research, catalog no. R1018) and ~35 μg fragmented total RNAs were then incubated with 15 μg anti-Pseudouridine antibody (MBL Life Science, catalog no. D347-3) or control mouse IgG (R&D Systems, catalog no. MAB002) in 1x RNA immunoprecipitation (IP) buffer (10 mM Tris-HCl pH 7.5, 250 mM NaCl, and 0.5% (vol/vol) Igepal CA-630) supplemented with 1 U/μl RNase inhibitor. The RNA/antibody mixtures were head-over-tail rotated for 2 hours at 4°C. Meanwhile, 100 μl Protein A+G magnetic beads (ThermoFisher, catalog no. 88803) were washed three times with 1 x RNA IP buffer and blocked in 0.5 mg/ml BSA in 1 x RNA IP buffer supplemented with 1 U/μl RNase inhibitor with head-over-tail rotation for 2 hours at 4°C. Next, the RNA/antibody mixtures were transferred into the bead-containing tubes prepared, and the reaction mixtures were incubated with head-over-tail rotation for 2 hours at 4°C. After incubation, the immunoprecipitated bead/antibody/RNA mixtures were washed three times with 1 x RNA IP buffer supplemented with 1 U/μl RNase inhibitor. The immunoprecipitated RNAs were then extracted using the Qiagen RNeasy micro kit (Qiagen, catalog no. 74004) and subjected to qRT-PCR.

Statistical analysis

Quantitative densitometric analysis of immunoblotting and Northern blotting images was performed using BioRad ImageLab (version 6.1.0). Statistical analyses were performed using GraphPad Prism software (version 9). The data of immunoblotting, Northern blotting, and qRT-PCR is expressed as mean ± SEM, and their statistical significance was assessed by the two-tailed unpaired Student's t test. CellProfiler (version 4.2.5) was used to count LC3B puncta or fluorescent signals from immunofluorescent images, and the statistical significance was calculated using the Mann-Whitney test. For the transcriptomics data, differential expression analysis was performed using DESeq2 to generate Benjamini–Hochberg-corrected P values (padj) to assess the statistical significance. For proteomics data, protein abundances were normalized using variance stabilization normalization (vsn package) in R v3.5.2. Statistical significance in protein abundances between the control and tRNA-Asp-GTC-3′tDR pulldown experiment was assessed using a Student's t test. Proteins with P value <0.05 and $\log_2$ fold-change (tRNA-Asp-GTC-3′tDR/control) >1 were selected as significantly interacting proteins in the tRNA-Asp-GTC-3′tDR pulldown experiment. For patient characteristics data, the Mann-Whitney test was used to assess the statistical significance of means between the two groups. Fisher's exact test was used to compare the proportions between the two groups. The criterion for statistical significance was $P < 0.05$ ($^*P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$).

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ACKNOWLEDGMENTS

We thank D. E. Capen from the Microscopy Core of the Program in Membrane Biology at Massachusetts General Hospital for assistance with electron microscopy imaging; M. Song for assistance with the injection of the modified ASOs and tissue harvesting; L. Lamale-Smith, G. Ramos, M. Tarsa, R. Wolf, and K. Zhang-Rutledge for help with the adjudication of preeclampsia cases; M. Jacobs, D. Melber, and C. Wauer for help with the identification of cases and controls for the preeclampsia urine cohort and abstraction of

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metadata from the UC San Diego Center for Perinatal Discovery's Perinatal Biorepository and Database; and B. ElGhayati for preparation of urine small RNA libraries. **Funding:** G.L. is supported by grants from the American Heart Association (AHA grant 23CDA1045944) and the National Institutes of Health (NIH grant R21DK137432). C.Xin. is supported by the NIH (grants R21DK137432 and R01DK072381). H.L. is supported by the AHA (grants 20CDA35310184 and 24SCEFIA1253853) and the NIH (grant AG077040). P.Pa. is supported by the NIH (grant R00HD096125) and Preeclampsia Foundation Canada (Vision Grant 2021). P.Pi. is supported by the National Cancer Institute of the NIH (grant P30CA33572). L.C.L. is supported by the NIH (grant UG3CA241687). P.I. is supported by the NIH (grants R01 GM126150 and R01 GM146997). J.V.B. is supported by the NIH (grants R01DK39773 and R01DK072381). S.D. is supported the NIH (NHLBI grant R35HL150807, NCATS grant UG3TR002878, and NIDDK grant R21DK137432). **Author contributions:** Conceptualization: G.L., S.D. Data analysis: G.L., A.C.M., T.H., C.L.O., H.M., K.G., S.L., Q.H., P.K., P.I. Funding acquisition: S.D., G.L., J.B., C.Xin. Investigation: G.L., L.S., C.Xin, T.H., H.M., P.K., P.G., X.Y., R.S., P.Pantham, C.Xiao, H.Y.W., E.C., S.Y., L.B.R., M.S., H.L. Methodology: G.L., S.D., J.V.B., P.I., M.B., T.M.L. Supervision: S.D., J.V.B., M.B., G.L., P.I. Writing – original draft: G.L., S.D. Writing – review & editing: all authors. **Competing interests:** R.V.S. has equity ownership in Thryv Therapeutics. R.V.S. is a coinventor on pending patents or disclosures on molecular biomarkers of fitness, lung disease, cardiovascular diseases and phenotypes, and metabolic health and the use of RNAs as therapeutics and diagnostic biomarkers in disease. O.A. is a consultant and holds equity in Reversal Therapeutics. J.V.B. is an inventor on KIM-1 patents assigned to Mass General Brigham; is the editor of *Seminars in Nephrology*; is a consultant to Sarepta, Mitsubishi-Tanabe, GentiBio, and Praxis; and has equity interests in Renalytix, DxNow, Oisin, VeriNano, Autonomous, Medical Devices, Pacific Biosciences, Medibeacon, and Cascade Medical Systems. S.D. is a founding member and has equity and consulting agreements for Thryv Therapeutics and Switch Therapeutics, neither of which is relevant to this study. S.D. received research support from Abbott and Bristol Myers Squibb that was not relevant to this study. G.L. and S.D. are coinventors on a US provisional patent application (WO2024264035A2) relating to tsRNA/tDR generation and engineering technology filed by Mass General Brigham. The remaining authors declare no competing interests. **Data and materials availability:** The human kidney tissue scRNA-seq data were obtained from GSE183279 (https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE183279). The ARM-seq data of HEK cells treated with different stressors were obtained from GSE173806 (https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE173806). The long RNA-seq data and OTTR-seq data of HEK cells transfected with the control or tRNA-Asp-GTC-3′tDR mimics have been deposited in the Gene Expression Omnibus of the National Center for Biotechnology Information (accession numbers: GSE295814 and GSE289124). Proteomics data were deposited in PRIDE (accession number: PXD052235). Other materials are available from S.D. under a material transfer agreement with Mass General Brigham. **License information:** Copyright © 2025 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. https://www.science.org/about/science-licenses-journal-article-reuse

SUPPLEMENTARY MATERIALS

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Submitted 1 April 2024; resubmitted 7 February 2025; accepted 10 June 2025; published online 17 July 2025

10.1126/science.adp5384

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A hypoxia-responsive tRNA-derived small RNA confers renal protection through RNA autophagy

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Science **389** (6763), eadp5384. DOI: 10.1126/science.adp5384

Editor's summary

Transfer RNAs (tRNAs) can be cut into smaller pieces called tRNA-derived small RNAs (tDRs) that can sometimes help cells respond to stress. Li *et al*. found that levels of one such RNA, Asp-GTC-3#tDR, markedly increased in response to disease-related stress signals in kidney cell culture and mouse models of kidney diseases. Asp-GTC-3#tDR was protective in kidney cells and regulated autophagy, a process by which a cell can recycle its RNA and proteins. Inhibiting Asp-GTC-3#tDR in kidney disease models worsened renal injury, cell death, inflammation, and fibrosis, whereas increasing its abundance protected against these processes. Targeting of tDRs may thus represent a strategy for treating kidney disease. —Stella M. Hurtley

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